

Phase I/II Operating Characteristics

Shared vs dose-specific additive-alpha efficacy - IPDE alpha = 0.9 - allocation = two-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No MTD/OBD %
Scenario 1						
DLT rate	0.30	0.35	0.40	0.45	0.50	
Efficacy rate	0.05	0.06	0.07	0.08	0.09	
Utility 2	-0.2	-0.3	-0.3	-0.4	-0.4	
Utility 3	31	29.6	28.2	26.8	25.4	
MTD selection (%)						
U2 - shared alpha	71.4	6.6	1.3	0	0	20.7
U2 - dose-specific alpha	72	6.5	1.4	0	0	20.1
U3 - shared alpha	71.5	6.6	1.2	0	0	20.7
U3 - dose-specific alpha	71.4	6.6	1.6	0	0	20.4
OBD selection (%)						
U2 - shared alpha	11.5	2.8	0.2	0	0	85.5
U2 - dose-specific alpha	11.5	2.7	0.4	0	0	85.4
U3 - shared alpha	11.7	2.3	0.2	0	0	85.8
U3 - dose-specific alpha	12.3	1.9	0.4	0	0	85.4
Mean administrations						
U2 - shared alpha	11.7	2.4	1	0.2	0	-
U2 - dose-specific alpha	11.7	2.4	0.9	0.2	0	-
U3 - shared alpha	11.7	2.4	1	0.2	0	-
U3 - dose-specific alpha	11.8	2.4	0.9	0.2	0	-
Mean IPDE administrations						
U2 - shared alpha	3.5	1.3	0.4	0.1	0	-
U2 - dose-specific alpha	3.5	1.4	0.4	0.1	0	-
U3 - shared alpha	3.5	1.3	0.4	0.1	0	-
U3 - dose-specific alpha	3.5	1.4	0.4	0.1	0	-
Mean trial duration (days)						Days
U2 - shared alpha						543.2
U2 - dose-specific alpha						542.1
U3 - shared alpha						542.6
U3 - dose-specific alpha						543

Scenario 2						
DLT rate	0.15	0.20	0.30	0.35	0.45	
Efficacy rate	0.20	0.25	0.50	0.50	0.50	
Utility 2	0.1	0.1	0.2	0.1	0.1	
Utility 3	46	47	58	56	52	
MTD selection (%)						
U2 - shared alpha	31.1	27.9	28.5	10	1.7	0.8
U2 - dose-specific alpha	31.6	27	28.3	10.9	1.4	0.8
U3 - shared alpha	31.5	27.1	29	9.9	1.7	0.8
U3 - dose-specific alpha	31.6	27.2	28.5	10.1	1.8	0.8
OBD selection (%)						
U2 - shared alpha	42.1	22.5	27.3	5	0.5	2.6
U2 - dose-specific alpha	42.3	22.5	26.8	5.3	0.5	2.6
U3 - shared alpha	45	22.5	24.9	4.4	0.5	2.7
U3 - dose-specific alpha	44.9	23.9	24	4.1	0.4	2.7
Mean administrations						
U2 - shared alpha	13.6	6	6.3	2.7	0.8	-
U2 - dose-specific alpha	13.5	6.1	6.4	2.7	0.8	-
U3 - shared alpha	13.6	6	6.3	2.7	0.8	-
U3 - dose-specific alpha	13.6	6	6.3	2.7	0.9	-
Mean IPDE administrations						
U2 - shared alpha	4.1	2.8	1.8	0.7	0.2	-
U2 - dose-specific alpha	4.1	2.8	1.8	0.7	0.2	-
U3 - shared alpha	4.1	2.8	1.8	0.8	0.2	-
U3 - dose-specific alpha	4.1	2.8	1.8	0.7	0.2	-
Mean trial duration (days)						Days
U2 - shared alpha						1092.7
U2 - dose-specific alpha						1093.3
U3 - shared alpha						1093.2
U3 - dose-specific alpha						1092.6

Priors: efficacy beta-binomial Beta(0.5, 0.5); efficacy additive alpha Beta(0.15, 0.85); toxicity additive alpha Beta(0.15, 0.85).

Final column: early stopping % for MTD selection and no-OBD % for OBD selection; shaded rows are the dose-specific alpha efficacy model.

Phase I/II Operating Characteristics

Shared vs dose-specific additive-alpha efficacy - IPDE alpha = 0.9 - allocation = two-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No MTD/OBD %
Scenario 3						
DLT rate	0.15	0.20	0.30	0.35	0.45	
Efficacy rate	0.10	0.40	0.40	0.40	0.40	
Utility 2	-0.1	0.2	0.1	0.1	-0.1	
Utility 3	40	56	52	50	46	
MTD selection (%)						
U2 - shared alpha	32.8	27.1	28	9.7	1.5	0.9
U2 - dose-specific alpha	32.8	26.7	28.3	9.7	1.6	0.9
U3 - shared alpha	32.8	27.4	27.6	9.7	1.6	0.9
U3 - dose-specific alpha	33.1	26.9	28	9.5	1.7	0.8
OBD selection (%)						
U2 - shared alpha	19	41.3	17.3	3.6	0.5	18.3
U2 - dose-specific alpha	18.6	41.7	16.3	4.8	0.5	18.1
U3 - shared alpha	20.5	44.1	13.5	3.5	0.3	18.1
U3 - dose-specific alpha	19.9	44.4	13.7	3.5	0.4	18.1
Mean administrations						
U2 - shared alpha	10.4	8	5.3	2.5	0.8	-
U2 - dose-specific alpha	10.4	8	5.3	2.5	0.8	-
U3 - shared alpha	10.4	8.1	5.2	2.5	0.8	-
U3 - dose-specific alpha	10.5	8	5.3	2.4	0.8	-
Mean IPDE administrations						
U2 - shared alpha	3.1	3.4	1.5	0.8	0.2	-
U2 - dose-specific alpha	3.1	3.4	1.5	0.9	0.2	-
U3 - shared alpha	3.1	3.4	1.5	0.8	0.2	-
U3 - dose-specific alpha	3.1	3.4	1.5	0.8	0.3	-
Mean trial duration (days)						Days
U2 - shared alpha						987.2
U2 - dose-specific alpha						986.9
U3 - shared alpha						987.3
U3 - dose-specific alpha						987.3

Scenario 4						
DLT rate	0.05	0.10	0.18	0.30	0.40	
Efficacy rate	0.05	0.10	0.25	0.50	0.55	
Utility 2	0	0	0.1	0.2	0.1	
Utility 3	41	42	47.8	58	57	
MTD selection (%)						
U2 - shared alpha	3.6	6.3	32.8	43.5	13.8	0
U2 - dose-specific alpha	3.4	6.4	33.3	43.3	13.6	0
U3 - shared alpha	3.5	6.6	32.6	43.3	14	0
U3 - dose-specific alpha	3.6	5.7	33	43.7	14	0
OBD selection (%)						
U2 - shared alpha	7.4	12.3	27.2	40.5	8	4.6
U2 - dose-specific alpha	8.2	11.6	27.8	40.4	7.3	4.7
U3 - shared alpha	12.9	11.7	25.3	38.6	6.9	4.6
U3 - dose-specific alpha	12.3	12.8	24.6	39.3	6.6	4.4
Mean administrations						
U2 - shared alpha	4.6	4.8	7.5	7.8	4.5	-
U2 - dose-specific alpha	4.6	4.9	7.5	7.8	4.5	-
U3 - shared alpha	4.6	4.8	7.5	7.9	4.4	-
U3 - dose-specific alpha	4.7	4.8	7.5	7.8	4.5	-
Mean IPDE administrations						
U2 - shared alpha	0.7	3	2.4	2.6	1.1	-
U2 - dose-specific alpha	0.7	3	2.4	2.6	1.1	-
U3 - shared alpha	0.7	3	2.4	2.6	1.1	-
U3 - dose-specific alpha	0.8	3	2.4	2.6	1.1	-
Mean trial duration (days)						Days
U2 - shared alpha						1067.1
U2 - dose-specific alpha						1065.4
U3 - shared alpha						1065.5
U3 - dose-specific alpha						1064.8

Priors: efficacy beta-binomial Beta(0.5, 0.5); efficacy additive alpha Beta(0.15, 0.85); toxicity additive alpha Beta(0.15, 0.85).

Final column: early stopping % for MTD selection and no-OBD % for OBD selection; shaded rows are the dose-specific alpha efficacy model.

Phase I/II Operating Characteristics

Shared vs dose-specific additive-alpha efficacy - IPDE alpha = 0.9 - allocation = two-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No MTD/OBD %
Scenario 5						
DLT rate	0.07	0.12	0.17	0.22	0.30	
Efficacy rate	0.05	0.30	0.30	0.30	0.30	
Utility 2	-0	0.2	0.1	0.1	0	
Utility 3	40.2	53.2	51.2	49.2	46	
MTD selection (%)						
U2 - shared alpha	8.3	7.5	18.5	33.9	31.7	0.1
U2 - dose-specific alpha	8.3	7.6	18.9	32.7	32.4	0.1
U3 - shared alpha	8.3	7.6	18.9	32.2	32.9	0.1
U3 - dose-specific alpha	8.3	7.3	19	32.8	32.5	0.1
OBD selection (%)						
U2 - shared alpha	3.6	35.1	26.6	19.6	7	8.1
U2 - dose-specific alpha	3.8	36.1	25.6	19.6	6.8	8.1
U3 - shared alpha	6.1	40.8	24.2	16.1	4.7	8.1
U3 - dose-specific alpha	5.3	40.2	24.9	17.6	3.9	8.1
Mean administrations						
U2 - shared alpha	4.8	6.8	6.9	5.8	4.3	-
U2 - dose-specific alpha	4.8	6.8	6.8	5.8	4.2	-
U3 - shared alpha	4.8	6.8	6.9	5.8	4.3	-
U3 - dose-specific alpha	4.8	6.7	6.9	5.8	4.3	-
Mean IPDE administrations						
U2 - shared alpha	0.8	3.6	2.2	2.2	1.5	-
U2 - dose-specific alpha	0.8	3.6	2.1	2.2	1.5	-
U3 - shared alpha	0.8	3.6	2.2	2.2	1.6	-
U3 - dose-specific alpha	0.8	3.6	2.2	2.2	1.5	-
Mean trial duration (days)						Days
U2 - shared alpha						1000.9
U2 - dose-specific alpha						1000.2
U3 - shared alpha						1001
U3 - dose-specific alpha						1000.4

Scenario 6						
DLT rate	0.29	0.34	0.38	0.43	0.52	
Efficacy rate	0.30	0.32	0.35	0.38	0.40	
Utility 2	0	-0	-0	-0.1	-0.1	
Utility 3	46.4	45.6	45.8	45.6	43.2	
MTD selection (%)						
U2 - shared alpha	61.8	10.5	2	0.5	0	25.2
U2 - dose-specific alpha	62.7	10.4	1.9	0.5	0	24.5
U3 - shared alpha	62	10.5	2.2	0.5	0	24.8
U3 - dose-specific alpha	62.1	10.4	2	0.5	0	25
OBD selection (%)						
U2 - shared alpha	64.6	6.5	1.3	0.1	0	27.5
U2 - dose-specific alpha	65.4	6.3	1	0.2	0	27.1
U3 - shared alpha	66	6	0.8	0.2	0	27
U3 - dose-specific alpha	65.8	5.8	0.8	0.2	0	27.4
Mean administrations						
U2 - shared alpha	20.5	3.2	1.4	0.3	0	-
U2 - dose-specific alpha	20.4	3.2	1.4	0.3	0	-
U3 - shared alpha	20.5	3.2	1.4	0.3	0	-
U3 - dose-specific alpha	20.4	3.2	1.4	0.3	0	-
Mean IPDE administrations						
U2 - shared alpha	5.7	1.3	0.4	0.1	0	-
U2 - dose-specific alpha	5.7	1.3	0.4	0.1	0	-
U3 - shared alpha	5.7	1.3	0.4	0.1	0	-
U3 - dose-specific alpha	5.7	1.3	0.4	0.1	0	-
Mean trial duration (days)						Days
U2 - shared alpha						978.2
U2 - dose-specific alpha						977.2
U3 - shared alpha						979.3
U3 - dose-specific alpha						978.5

Priors: efficacy beta-binomial Beta(0.5, 0.5); efficacy additive alpha Beta(0.15, 0.85); toxicity additive alpha Beta(0.15, 0.85).

Final column: early stopping % for MTD selection and no-OBD % for OBD selection; shaded rows are the dose-specific alpha efficacy model.

Phase I/II Operating Characteristics

Shared vs dose-specific additive-alpha efficacy - IPDE alpha = 0.9 - allocation = two-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No MTD/OBD %
Scenario 7						
DLT rate	0.16	0.20	0.26	0.31	0.36	
Efficacy rate	0.20	0.50	0.50	0.50	0.50	
Utility 2	0	0.3	0.2	0.2	0.1	
Utility 3	45.6	62	59.6	57.6	55.6	
MTD selection (%)						
U2 - shared alpha	33.3	22.1	24.1	15	4.4	1.1
U2 - dose-specific alpha	33.1	22	24.2	15.3	4.3	1.1
U3 - shared alpha	33.1	22	24.6	14.8	4.4	1.1
U3 - dose-specific alpha	32.7	23.1	23.8	15.2	4.1	1.1
OBD selection (%)						
U2 - shared alpha	35.4	39.2	15.9	4.8	1.3	3.4
U2 - dose-specific alpha	34.7	40.6	14.4	5.5	1.2	3.6
U3 - shared alpha	36	41.6	13.4	4.8	0.7	3.5
U3 - dose-specific alpha	34.7	44.5	11.6	4.7	0.8	3.7
Mean administrations						
U2 - shared alpha	13	7.5	4.9	2.7	1.3	-
U2 - dose-specific alpha	12.9	7.5	5	2.6	1.2	-
U3 - shared alpha	13	7.5	5.1	2.6	1.2	-
U3 - dose-specific alpha	13	7.5	5	2.6	1.2	-
Mean IPDE administrations						
U2 - shared alpha	3.8	3	1.2	0.8	0.4	-
U2 - dose-specific alpha	3.8	3	1.2	0.8	0.4	-
U3 - shared alpha	3.8	3	1.3	0.8	0.4	-
U3 - dose-specific alpha	3.8	3	1.2	0.8	0.4	-
Mean trial duration (days)						Days
U2 - shared alpha						1110.4
U2 - dose-specific alpha						1109.5
U3 - shared alpha						1110.2
U3 - dose-specific alpha						1109.6

Scenario 8						
DLT rate	0.32	0.37	0.42	0.48	0.52	
Efficacy rate	0.10	0.15	0.20	0.30	0.40	
Utility 2	-0.2	-0.2	-0.2	-0.2	-0.1	
Utility 3	33.2	34.2	35.2	38.8	43.2	
MTD selection (%)						
U2 - shared alpha	62.3	3.8	0.9	0	0	33
U2 - dose-specific alpha	63.7	3.7	0.9	0	0	31.7
U3 - shared alpha	62.5	3.8	0.9	0	0	32.8
U3 - dose-specific alpha	62.8	3.8	0.8	0	0	32.6
OBD selection (%)						
U2 - shared alpha	26.9	2.3	0.6	0	0	70.2
U2 - dose-specific alpha	27.3	2.4	0.6	0	0	69.7
U3 - shared alpha	27.2	2.3	0.4	0	0	70.1
U3 - dose-specific alpha	27.3	2.1	0.4	0	0	70.2
Mean administrations						
U2 - shared alpha	14.9	2.3	0.8	0.1	0	-
U2 - dose-specific alpha	14.9	2.3	0.8	0.1	0	-
U3 - shared alpha	14.9	2.3	0.8	0.1	0	-
U3 - dose-specific alpha	14.9	2.3	0.8	0.1	0	-
Mean IPDE administrations						
U2 - shared alpha	4.6	1.2	0.3	0.1	0	-
U2 - dose-specific alpha	4.6	1.2	0.3	0.1	0	-
U3 - shared alpha	4.6	1.2	0.3	0.1	0	-
U3 - dose-specific alpha	4.6	1.2	0.3	0.1	0	-
Mean trial duration (days)						Days
U2 - shared alpha						656.6
U2 - dose-specific alpha						657.1
U3 - shared alpha						656.6
U3 - dose-specific alpha						656.3

Priors: efficacy beta-binomial Beta(0.5, 0.5); efficacy additive alpha Beta(0.15, 0.85); toxicity additive alpha Beta(0.15, 0.85).

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Phase I/II Operating Characteristics

Shared vs dose-specific additive-alpha efficacy - IPDE alpha = 0.9 - allocation = two-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No MTD/OBD %
Scenario 9						
DLT rate	0.15	0.18	0.22	0.26	0.29	
Efficacy rate	0.05	0.08	0.20	0.25	0.50	
Utility 2	-0.1	-0.1	-0	-0	0.2	
Utility 3	37	37.6	43.2	44.6	58.4	
MTD selection (%)						
U2 - shared alpha	32.3	17.8	19.6	15.9	13.4	1
U2 - dose-specific alpha	32.3	17.2	20.4	16.2	12.9	1
U3 - shared alpha	32.2	17.1	20.7	15.9	13.1	1
U3 - dose-specific alpha	32.3	17.6	19.4	16.4	13.3	1
OBD selection (%)						
U2 - shared alpha	12.5	13.9	17.2	9.9	11.2	35.3
U2 - dose-specific alpha	14.6	12	17.6	9.9	10.9	35
U3 - shared alpha	16.8	13.3	16.2	9	9.3	35.4
U3 - dose-specific alpha	17.3	12.8	15.3	9.7	9.5	35.4
Mean administrations						
U2 - shared alpha	8.5	5	5.1	3.1	2.6	-
U2 - dose-specific alpha	8.4	5	5.1	3.1	2.6	-
U3 - shared alpha	8.5	4.9	5.1	3.1	2.6	-
U3 - dose-specific alpha	8.4	5	5.1	3.1	2.7	-
Mean IPDE administrations						
U2 - shared alpha	2.3	2.8	1.8	1.3	0.8	-
U2 - dose-specific alpha	2.3	2.8	1.8	1.3	0.8	-
U3 - shared alpha	2.3	2.8	1.8	1.3	0.8	-
U3 - dose-specific alpha	2.3	2.8	1.8	1.3	0.8	-
Mean trial duration (days)						Days
U2 - shared alpha						837.8
U2 - dose-specific alpha						838.3
U3 - shared alpha						837.2
U3 - dose-specific alpha						838.2

Scenario 10						
DLT rate	0.23	0.28	0.32	0.39	0.43	
Efficacy rate	0.05	0.40	0.42	0.45	0.50	
Utility 2	-0.2	0.1	0.1	0.1	0.1	
Utility 3	33.8	52.8	52.4	51.4	52.8	
MTD selection (%)						
U2 - shared alpha	62.2	18.4	9.1	2.7	0.6	7
U2 - dose-specific alpha	62.4	18.9	8.3	3.1	0.5	6.8
U3 - shared alpha	62.2	18.7	8.5	2.9	0.6	7.1
U3 - dose-specific alpha	62.2	19	8.5	2.8	0.5	7
OBD selection (%)						
U2 - shared alpha	11	20.9	6.7	1.5	0.4	59.5
U2 - dose-specific alpha	11	21.9	5.7	1.4	0.4	59.6
U3 - shared alpha	11.6	21.2	5.8	1.6	0.3	59.5
U3 - dose-specific alpha	11.2	22.4	5.1	1.4	0.2	59.7
Mean administrations						
U2 - shared alpha	10.4	5.1	2.9	1	0.3	-
U2 - dose-specific alpha	10.3	5.2	2.8	1	0.3	-
U3 - shared alpha	10.3	5.2	2.9	1	0.2	-
U3 - dose-specific alpha	10.4	5.1	2.9	0.9	0.3	-
Mean IPDE administrations						
U2 - shared alpha	3	2.3	0.8	0.3	0.1	-
U2 - dose-specific alpha	2.9	2.3	0.8	0.3	0.1	-
U3 - shared alpha	2.9	2.3	0.8	0.3	0.1	-
U3 - dose-specific alpha	3	2.3	0.8	0.3	0.1	-
Mean trial duration (days)						Days
U2 - shared alpha						714.5
U2 - dose-specific alpha						714.8
U3 - shared alpha						714.8
U3 - dose-specific alpha						714.6

Priors: efficacy beta-binomial Beta(0.5, 0.5); efficacy additive alpha Beta(0.15, 0.85); toxicity additive alpha Beta(0.15, 0.85).

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Shared vs dose-specific additive-alpha efficacy - IPDE alpha = 0.9 - allocation = two-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No MTD/OBD %
Scenario 11						
DLT rate	0.22	0.32	0.44	0.59	0.67	
Efficacy rate	0.20	0.20	0.20	0.20	0.20	
Utility 2	-0	-0.1	-0.2	-0.4	-0.5	
Utility 3	43.2	39.2	34.4	28.4	25.2	
MTD selection (%)						
U2 - shared alpha	67.3	23.6	2.5	0.1	0	6.5
U2 - dose-specific alpha	66.7	23.4	2.8	0.2	0	6.9
U3 - shared alpha	67.3	22.9	2.8	0.2	0	6.8
U3 - dose-specific alpha	67.5	23.3	2.1	0.2	0	6.9
OBD selection (%)						
U2 - shared alpha	72.1	13.4	0.5	0.1	0	13.9
U2 - dose-specific alpha	71.4	13.2	1	0.2	0	14.2
U3 - shared alpha	73	12.6	0.5	0.1	0	13.8
U3 - dose-specific alpha	73	12.7	0.4	0.1	0	13.8
Mean administrations						
U2 - shared alpha	20.3	5.2	1.9	0.3	0	-
U2 - dose-specific alpha	20.2	5.2	2	0.3	0	-
U3 - shared alpha	20.3	5.2	1.9	0.3	0	-
U3 - dose-specific alpha	20.2	5.2	2	0.3	0	-
Mean IPDE administrations						
U2 - shared alpha	6.4	2.4	0.7	0.1	0	-
U2 - dose-specific alpha	6.4	2.4	0.7	0.1	0	-
U3 - shared alpha	6.4	2.4	0.7	0.1	0	-
U3 - dose-specific alpha	6.4	2.4	0.7	0.1	0	-
Mean trial duration (days)						Days
U2 - shared alpha						995.1
U2 - dose-specific alpha						994.3
U3 - shared alpha						995.5
U3 - dose-specific alpha						996

Scenario 12						
DLT rate	0.06	0.09	0.18	0.29	0.44	
Efficacy rate	0.05	0.10	0.12	0.40	0.60	
Utility 2	-0	0	-0.1	0.1	0.2	
Utility 3	40.6	42.4	40	52.4	58.4	
MTD selection (%)						
U2 - shared alpha	5.3	5.2	35.3	41.5	12.6	0.1
U2 - dose-specific alpha	5.5	5.3	35	42.1	12	0.1
U3 - shared alpha	5.3	5.2	35.3	42	12.1	0.1
U3 - dose-specific alpha	5.5	5.3	34.2	42.6	12.3	0.1
OBD selection (%)						
U2 - shared alpha	16.3	17.9	15.5	34	9.9	6.4
U2 - dose-specific alpha	16.1	19.2	14.7	34.1	9.3	6.6
U3 - shared alpha	21.5	18.7	10.7	33.5	9.1	6.5
U3 - dose-specific alpha	22.4	18.2	10.4	33.1	9.1	6.8
Mean administrations						
U2 - shared alpha	5.2	5.5	6.3	7.3	4.7	-
U2 - dose-specific alpha	5.2	5.6	6.2	7.3	4.7	-
U3 - shared alpha	5.2	5.5	6.3	7.2	4.7	-
U3 - dose-specific alpha	5.2	5.5	6.2	7.2	4.7	-
Mean IPDE administrations						
U2 - shared alpha	1	3.3	2.1	2.7	1.1	-
U2 - dose-specific alpha	1	3.3	2.1	2.7	1.1	-
U3 - shared alpha	1	3.3	2.1	2.7	1.1	-
U3 - dose-specific alpha	1	3.3	2	2.7	1.1	-
Mean trial duration (days)						Days
U2 - shared alpha						1030.6
U2 - dose-specific alpha						1031.9
U3 - shared alpha						1030.9
U3 - dose-specific alpha						1031.3

Priors: efficacy beta-binomial Beta(0.5, 0.5); efficacy additive alpha Beta(0.15, 0.85); toxicity additive alpha Beta(0.15, 0.85).

Final column: early stopping % for MTD selection and no-OBD % for OBD selection; shaded rows are the dose-specific alpha efficacy model.

Phase I/II Operating Characteristics

Shared vs dose-specific additive-alpha efficacy - IPDE alpha = 0.9 - allocation = two-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No MTD/OBD %
Scenario 13						
DLT rate	0.21	0.29	0.30	0.39	0.46	
Efficacy rate	0.05	0.30	0.32	0.35	0.40	
Utility 2	-0.2	0	0	-0	-0.1	
Utility 3	34.6	46.4	47.2	45.4	45.6	
MTD selection (%)						
U2 - shared alpha	60.3	22.1	9.7	2.9	0.3	4.7
U2 - dose-specific alpha	60.5	21	10.5	2.9	0.3	4.8
U3 - shared alpha	59.9	22.1	10.4	2.4	0.4	4.8
U3 - dose-specific alpha	60	21.7	10.5	2.7	0.3	4.8
OBD selection (%)						
U2 - shared alpha	13.6	23.6	6.4	1.2	0.3	54.9
U2 - dose-specific alpha	13.8	22.8	7.3	1.3	0.2	54.6
U3 - shared alpha	14.8	23.5	6.1	1	0.3	54.3
U3 - dose-specific alpha	14.9	22.7	6.4	1	0.2	54.8
Mean administrations						
U2 - shared alpha	10.4	5.7	3.1	1.1	0.3	-
U2 - dose-specific alpha	10.3	5.6	3.2	1.1	0.3	-
U3 - shared alpha	10.3	5.6	3.1	1.1	0.3	-
U3 - dose-specific alpha	10.3	5.6	3.2	1.1	0.3	-
Mean IPDE administrations						
U2 - shared alpha	3	2.7	1	0.4	0.1	-
U2 - dose-specific alpha	2.9	2.7	1	0.4	0.1	-
U3 - shared alpha	3	2.7	1	0.4	0.1	-
U3 - dose-specific alpha	3	2.7	1	0.5	0.1	-
Mean trial duration (days)						Days
U2 - shared alpha						733
U2 - dose-specific alpha						731.8
U3 - shared alpha						730.9
U3 - dose-specific alpha						730.7

Scenario 14						
DLT rate	0.09	0.15	0.23	0.28	0.39	
Efficacy rate	0.05	0.08	0.12	0.30	0.40	
Utility 2	-0	-0.1	-0.1	0	0	
Utility 3	39.4	38.8	38	46.8	48.4	
MTD selection (%)						
U2 - shared alpha	14.6	14.5	33.6	28.1	9	0.2
U2 - dose-specific alpha	14.4	14.6	33.8	27.7	9.3	0.2
U3 - shared alpha	14.8	15	33.2	27.3	9.5	0.2
U3 - dose-specific alpha	14.5	14.9	33.9	26.9	9.6	0.2
OBD selection (%)						
U2 - shared alpha	20.1	20.3	16.9	18.9	6	17.8
U2 - dose-specific alpha	20.9	19.6	15.9	20.5	5.3	17.8
U3 - shared alpha	28	20.2	11.3	17.7	4.9	17.9
U3 - dose-specific alpha	27.9	18.8	12.9	17.6	4.7	18.1
Mean administrations						
U2 - shared alpha	7	6	6.3	5	3	-
U2 - dose-specific alpha	7	6	6.3	5	2.9	-
U3 - shared alpha	7	6	6.3	5	3	-
U3 - dose-specific alpha	7.1	6	6.2	4.9	3	-
Mean IPDE administrations						
U2 - shared alpha	1.8	3.4	2.2	2	0.9	-
U2 - dose-specific alpha	1.8	3.4	2.2	2	0.9	-
U3 - shared alpha	1.8	3.4	2.2	2	0.9	-
U3 - dose-specific alpha	1.8	3.4	2.2	2	0.9	-
Mean trial duration (days)						Days
U2 - shared alpha						930.3
U2 - dose-specific alpha						931.1
U3 - shared alpha						931.3
U3 - dose-specific alpha						930.9

Priors: efficacy beta-binomial Beta(0.5, 0.5); efficacy additive alpha Beta(0.15, 0.85); toxicity additive alpha Beta(0.15, 0.85).

Final column: early stopping % for MTD selection and no-OBD % for OBD selection; shaded rows are the dose-specific alpha efficacy model.

Phase I/II Operating Characteristics

Shared vs dose-specific additive-alpha efficacy - IPDE alpha = 0.9 - allocation = two-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No MTD/OBD %
Scenario 15						
DLT rate	0.13	0.20	0.32	0.39	0.46	
Efficacy rate	0.10	0.10	0.10	0.10	0.10	
Utility 2	-0	-0.1	-0.2	-0.3	-0.4	
Utility 3	40.8	38	33.2	30.4	27.6	
MTD selection (%)						
U2 - shared alpha	27.2	36.1	29	6.3	0.6	0.8
U2 - dose-specific alpha	27.2	35.1	29.7	6.6	0.6	0.8
U3 - shared alpha	27.4	34.9	29.5	6.6	0.8	0.8
U3 - dose-specific alpha	27.3	35.1	28.9	7.4	0.5	0.8
OBD selection (%)						
U2 - shared alpha	44.9	26.9	7.9	1.6	0.1	18.6
U2 - dose-specific alpha	44.5	27.5	8.1	1.4	0	18.5
U3 - shared alpha	50.8	24.9	4.4	1.4	0.1	18.4
U3 - dose-specific alpha	50	26.1	4.6	1	0	18.3
Mean administrations						
U2 - shared alpha	12	7.6	5.1	1.9	0.6	-
U2 - dose-specific alpha	12	7.6	5.1	1.9	0.6	-
U3 - shared alpha	12.1	7.6	5.1	2	0.6	-
U3 - dose-specific alpha	12	7.7	5	1.9	0.6	-
Mean IPDE administrations						
U2 - shared alpha	3.8	3.8	1.8	0.9	0.2	-
U2 - dose-specific alpha	3.8	3.8	1.8	0.9	0.2	-
U3 - shared alpha	3.8	3.8	1.8	0.9	0.2	-
U3 - dose-specific alpha	3.8	3.9	1.8	0.9	0.2	-
Mean trial duration (days)						Days
U2 - shared alpha						915.8
U2 - dose-specific alpha						916.1
U3 - shared alpha						915.5
U3 - dose-specific alpha						915.8

Scenario 16						
DLT rate	0.02	0.06	0.09	0.14	0.26	
Efficacy rate	0.20	0.20	0.20	0.20	0.20	
Utility 2	0.2	0.1	0.1	0.1	-0.1	
Utility 3	51.2	49.6	48.4	46.4	41.6	
MTD selection (%)						
U2 - shared alpha	0.8	0.9	4.1	16.2	78	0
U2 - dose-specific alpha	0.8	0.9	3.9	16.1	78.3	0
U3 - shared alpha	0.8	0.9	4.1	15.7	78.5	0
U3 - dose-specific alpha	0.8	0.9	4.2	15.3	78.8	0
OBD selection (%)						
U2 - shared alpha	23.9	33	18.5	14.4	10.2	0
U2 - dose-specific alpha	26.4	30.6	17.7	15.1	10.2	0
U3 - shared alpha	31.8	36	14.9	10.9	6.4	0
U3 - dose-specific alpha	31.9	36.1	13.8	11.4	6.8	0
Mean administrations						
U2 - shared alpha	5.6	6.2	5.5	5.9	6.8	-
U2 - dose-specific alpha	5.6	6.1	5.5	6	6.8	-
U3 - shared alpha	5.6	6.1	5.5	6	6.8	-
U3 - dose-specific alpha	5.6	6.1	5.5	5.9	6.9	-
Mean IPDE administrations						
U2 - shared alpha	1.1	3.4	1.9	2.6	2.7	-
U2 - dose-specific alpha	1.1	3.3	1.9	2.7	2.6	-
U3 - shared alpha	1.1	3.4	1.9	2.7	2.6	-
U3 - dose-specific alpha	1.1	3.4	1.9	2.7	2.7	-
Mean trial duration (days)						Days
U2 - shared alpha						1007.3
U2 - dose-specific alpha						1006.6
U3 - shared alpha						1007.7
U3 - dose-specific alpha						1006.6

Priors: efficacy beta-binomial Beta(0.5, 0.5); efficacy additive alpha Beta(0.15, 0.85); toxicity additive alpha Beta(0.15, 0.85).

Final column: early stopping % for MTD selection and no-OBD % for OBD selection; shaded rows are the dose-specific alpha efficacy model.

Phase I/II Operating Characteristics

Shared vs dose-specific additive-alpha efficacy - IPDE alpha = 0.9 - allocation = two-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No MTD/OBD %
Scenario 17						
DLT rate	0.11	0.23	0.39	0.58	0.76	
Efficacy rate	0.30	0.50	0.50	0.50	0.50	
Utility 2	0.2	0.3	0.1	-0.1	-0.3	
Utility 3	53.6	60.8	54.4	46.8	39.6	
MTD selection (%)						
U2 - shared alpha	24.4	49	25.1	1.3	0	0.2
U2 - dose-specific alpha	24.6	49.6	24.3	1.3	0	0.2
U3 - shared alpha	24.4	49.4	24.9	1.1	0	0.2
U3 - dose-specific alpha	25.1	48.4	25	1.3	0	0.2
OBD selection (%)						
U2 - shared alpha	34	55.2	10	0.3	0	0.5
U2 - dose-specific alpha	35.1	55.6	8.8	0.1	0	0.4
U3 - shared alpha	36.6	54.9	8	0.1	0	0.4
U3 - dose-specific alpha	37.2	54.7	7.6	0.1	0	0.4
Mean administrations						
U2 - shared alpha	11.7	10.6	6.2	1.2	0.1	-
U2 - dose-specific alpha	11.7	10.8	6	1.2	0.2	-
U3 - shared alpha	11.7	10.7	6.1	1.2	0.1	-
U3 - dose-specific alpha	11.7	10.7	6	1.3	0.1	-
Mean IPDE administrations						
U2 - shared alpha	3.1	3.6	1.4	0.4	0	-
U2 - dose-specific alpha	3.1	3.7	1.4	0.4	0	-
U3 - shared alpha	3.1	3.7	1.4	0.4	0	-
U3 - dose-specific alpha	3.1	3.7	1.4	0.4	0	-
Mean trial duration (days)						Days
U2 - shared alpha						1177.3
U2 - dose-specific alpha						1178
U3 - shared alpha						1177.8
U3 - dose-specific alpha						1178.3

Scenario 18						
DLT rate	0.02	0.04	0.10	0.20	0.35	
Efficacy rate	0.05	0.10	0.15	0.20	0.50	
Utility 2	0	0.1	0.1	0	0.1	
Utility 3	42.2	44.4	45	44	56	
MTD selection (%)						
U2 - shared alpha	0.8	0.5	6	41.1	51.6	0
U2 - dose-specific alpha	0.8	0.4	5.9	42.2	50.7	0
U3 - shared alpha	0.8	0.4	6.3	41.8	50.7	0
U3 - dose-specific alpha	0.8	0.4	5.9	42.2	50.7	0
OBD selection (%)						
U2 - shared alpha	9	14.3	12.5	19.9	43.4	0.9
U2 - dose-specific alpha	9.9	13.3	12.8	20.4	42.7	0.9
U3 - shared alpha	14	15.9	13.4	17.5	38.3	0.9
U3 - dose-specific alpha	14.2	16.8	11.8	18.9	37.5	0.8
Mean administrations						
U2 - shared alpha	3.7	4.1	5	6	11.1	-
U2 - dose-specific alpha	3.7	4.1	5	6	10.9	-
U3 - shared alpha	3.7	4.1	5	6	11	-
U3 - dose-specific alpha	3.7	4.1	5	6.1	11	-
Mean IPDE administrations						
U2 - shared alpha	0.3	2.8	1.6	2.6	3.1	-
U2 - dose-specific alpha	0.3	2.8	1.7	2.7	3.1	-
U3 - shared alpha	0.3	2.8	1.7	2.7	3.1	-
U3 - dose-specific alpha	0.3	2.8	1.7	2.7	3.1	-
Mean trial duration (days)						Days
U2 - shared alpha						1063
U2 - dose-specific alpha						1062.3
U3 - shared alpha						1062.5
U3 - dose-specific alpha						1062.2

Priors: efficacy beta-binomial Beta(0.5, 0.5); efficacy additive alpha Beta(0.15, 0.85); toxicity additive alpha Beta(0.15, 0.85).

Final column: early stopping % for MTD selection and no-OBD % for OBD selection; shaded rows are the dose-specific alpha efficacy model.

Phase I/II Operating Characteristics

Shared vs dose-specific additive-alpha efficacy - IPDE alpha = 0.9 - allocation = two-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No MTD/OBD %
Scenario 19						
DLT rate	0.11	0.26	0.39	0.60	0.77	
Efficacy rate	0.08	0.30	0.32	0.35	0.40	
Utility 2	-0	0	-0.1	-0.2	-0.4	
Utility 3	40.4	47.6	43.6	37	33.2	
MTD selection (%)						
U2 - shared alpha	35.4	47.9	15.9	0.5	0	0.3
U2 - dose-specific alpha	35.5	48.2	15.5	0.5	0	0.3
U3 - shared alpha	35.3	48.6	15.4	0.4	0	0.3
U3 - dose-specific alpha	35.3	48.2	15.8	0.4	0	0.3
OBD selection (%)						
U2 - shared alpha	25.1	49.1	7.2	0.3	0	18.3
U2 - dose-specific alpha	26	48.4	6.9	0.3	0	18.4
U3 - shared alpha	26.9	47.9	6.3	0.3	0	18.6
U3 - dose-specific alpha	27	48.2	6.3	0.3	0	18.2
Mean administrations						
U2 - shared alpha	9.9	10.7	5.3	1	0.1	-
U2 - dose-specific alpha	10	10.6	5.3	1	0.1	-
U3 - shared alpha	9.9	10.6	5.3	1	0.1	-
U3 - dose-specific alpha	10	10.6	5.3	1	0.1	-
Mean IPDE administrations						
U2 - shared alpha	2.8	4.6	1.6	0.4	0	-
U2 - dose-specific alpha	2.9	4.6	1.6	0.4	0	-
U3 - shared alpha	2.8	4.6	1.6	0.4	0	-
U3 - dose-specific alpha	2.9	4.6	1.6	0.4	0	-
Mean trial duration (days)						Days
U2 - shared alpha						964.5
U2 - dose-specific alpha						963
U3 - shared alpha						963.2
U3 - dose-specific alpha						962.6

Scenario 20						
DLT rate	0.07	0.14	0.31	0.41	0.52	
Efficacy rate	0.05	0.25	0.25	0.25	0.25	
Utility 2	-0	0.1	-0.1	-0.2	-0.3	
Utility 3	40.2	49.4	42.6	38.6	34.2	
MTD selection (%)						
U2 - shared alpha	10.3	26	50.9	11.6	1.1	0.1
U2 - dose-specific alpha	10.1	26	51.1	11.4	1.3	0.1
U3 - shared alpha	10.3	25.8	50.5	12.1	1.2	0.1
U3 - dose-specific alpha	10.1	26.2	50.7	11.6	1.3	0.1
OBD selection (%)						
U2 - shared alpha	10.7	50.6	25	4.4	0	9.3
U2 - dose-specific alpha	11.1	51.8	24.2	3.7	0.2	9
U3 - shared alpha	14	51.1	22	3.8	0.1	9
U3 - dose-specific alpha	14.5	50.8	22.4	3.3	0.2	8.8
Mean administrations						
U2 - shared alpha	5.8	9.8	8.4	3.3	1	-
U2 - dose-specific alpha	5.7	9.9	8.5	3.2	1	-
U3 - shared alpha	5.7	9.8	8.6	3.3	1	-
U3 - dose-specific alpha	5.8	9.8	8.5	3.2	1	-
Mean IPDE administrations						
U2 - shared alpha	1.2	4.7	2.6	1.3	0.4	-
U2 - dose-specific alpha	1.2	4.8	2.6	1.3	0.4	-
U3 - shared alpha	1.2	4.8	2.7	1.3	0.4	-
U3 - dose-specific alpha	1.2	4.7	2.7	1.3	0.4	-
Mean trial duration (days)						Days
U2 - shared alpha						993.2
U2 - dose-specific alpha						992.8
U3 - shared alpha						994
U3 - dose-specific alpha						993.8

Priors: efficacy beta-binomial Beta(0.5, 0.5); efficacy additive alpha Beta(0.15, 0.85); toxicity additive alpha Beta(0.15, 0.85).

Final column: early stopping % for MTD selection and no-OBD % for OBD selection; shaded rows are the dose-specific alpha efficacy model.

Phase I/II Operating Characteristics

Shared vs dose-specific additive-alpha efficacy - IPDE alpha = 0.9 - allocation = two-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No MTD/OBD %
Scenario 21						
DLT rate	0.37	0.56	0.69	0.83	0.88	
Efficacy rate	0.50	0.50	0.50	0.50	0.50	
Utility 2	0.1	-0.1	-0.2	-0.3	-0.4	
Utility 3	55.2	47.6	42.4	36.8	34.8	
MTD selection (%)						
U2 - shared alpha	40.1	0.4	0	0	0	59.5
U2 - dose-specific alpha	40.6	0.5	0	0	0	58.9
U3 - shared alpha	39.9	0.4	0	0	0	59.7
U3 - dose-specific alpha	40	0.4	0	0	0	59.6
OBD selection (%)						
U2 - shared alpha	37.4	0.2	0	0	0	62.4
U2 - dose-specific alpha	38.1	0.2	0	0	0	61.7
U3 - shared alpha	37.8	0.2	0	0	0	62
U3 - dose-specific alpha	37.8	0.2	0	0	0	62
Mean administrations						
U2 - shared alpha	18.7	1.4	0.2	0	0	-
U2 - dose-specific alpha	18.7	1.4	0.2	0	0	-
U3 - shared alpha	18.8	1.4	0.2	0	0	-
U3 - dose-specific alpha	18.7	1.4	0.2	0	0	-
Mean IPDE administrations						
U2 - shared alpha	3.8	0.5	0.1	0	0	-
U2 - dose-specific alpha	3.8	0.5	0.1	0	0	-
U3 - shared alpha	3.8	0.5	0.1	0	0	-
U3 - dose-specific alpha	3.8	0.5	0.1	0	0	-
Mean trial duration (days)						Days
U2 - shared alpha						878.7
U2 - dose-specific alpha						880.5
U3 - shared alpha						882.7
U3 - dose-specific alpha						880.4

Scenario 22						
DLT rate	0.07	0.27	0.50	0.56	0.78	
Efficacy rate	0.08	0.25	0.25	0.50	0.50	
Utility 2	0	-0	-0.2	-0.1	-0.3	
Utility 3	42	44.2	35	47.6	38.8	
MTD selection (%)						
U2 - shared alpha	32.5	53.3	13.9	0.1	0	0.2
U2 - dose-specific alpha	32.5	52.8	14.4	0.1	0	0.2
U3 - shared alpha	32.5	53.2	14	0.1	0	0.2
U3 - dose-specific alpha	32.3	53	14.4	0.1	0	0.2
OBD selection (%)						
U2 - shared alpha	30.2	52	5.6	0.1	0	12.1
U2 - dose-specific alpha	30.6	51.3	5.8	0.1	0	12.2
U3 - shared alpha	33	49.8	5.1	0.1	0	12
U3 - dose-specific alpha	32	50.7	5.4	0.1	0	11.8
Mean administrations						
U2 - shared alpha	9.6	12.8	4.8	0.7	0	-
U2 - dose-specific alpha	9.6	12.8	4.8	0.7	0	-
U3 - shared alpha	9.5	12.8	4.8	0.7	0	-
U3 - dose-specific alpha	9.6	12.8	4.7	0.7	0	-
Mean IPDE administrations						
U2 - shared alpha	2.7	5.6	1.4	0.3	0	-
U2 - dose-specific alpha	2.7	5.6	1.4	0.3	0	-
U3 - shared alpha	2.7	5.6	1.4	0.3	0	-
U3 - dose-specific alpha	2.7	5.6	1.4	0.3	0	-
Mean trial duration (days)						Days
U2 - shared alpha						980.3
U2 - dose-specific alpha						980
U3 - shared alpha						981
U3 - dose-specific alpha						980.9

Priors: efficacy beta-binomial Beta(0.5, 0.5); efficacy additive alpha Beta(0.15, 0.85); toxicity additive alpha Beta(0.15, 0.85).

Final column: early stopping % for MTD selection and no-OBD % for OBD selection; shaded rows are the dose-specific alpha efficacy model.

Phase I/II Operating Characteristics

Shared vs dose-specific additive-alpha efficacy - IPDE alpha = 0.9 - allocation = two-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No MTD/OBD %
Scenario 23						
DLT rate	0.32	0.55	0.80	0.94	0.97	
Efficacy rate	0.05	0.05	0.05	0.05	0.05	
Utility 2	-0.3	-0.5	-0.8	-0.9	-0.9	
Utility 3	30.2	21	11	5.4	4.2	
MTD selection (%)						
U2 - shared alpha	70.6	1.1	0	0	0	28.3
U2 - dose-specific alpha	70.4	1	0	0	0	28.6
U3 - shared alpha	71.1	1	0	0	0	27.9
U3 - dose-specific alpha	70.9	1	0	0	0	28.1
OBD selection (%)						
U2 - shared alpha	5.6	1	0	0	0	93.4
U2 - dose-specific alpha	5.7	1	0	0	0	93.3
U3 - shared alpha	5.6	1	0	0	0	93.4
U3 - dose-specific alpha	5.7	1	0	0	0	93.3
Mean administrations						
U2 - shared alpha	11.8	1.4	0.2	0	0	-
U2 - dose-specific alpha	11.8	1.4	0.2	0	0	-
U3 - shared alpha	11.8	1.4	0.2	0	0	-
U3 - dose-specific alpha	11.8	1.4	0.2	0	0	-
Mean IPDE administrations						
U2 - shared alpha	3.5	0.9	0.1	0	0	-
U2 - dose-specific alpha	3.5	0.9	0.1	0	0	-
U3 - shared alpha	3.5	0.9	0.1	0	0	-
U3 - dose-specific alpha	3.5	0.9	0.1	0	0	-
Mean trial duration (days)						Days
U2 - shared alpha						477.3
U2 - dose-specific alpha						478.9
U3 - shared alpha						477.4
U3 - dose-specific alpha						478.3

Scenario 24						
DLT rate	0.01	0.02	0.05	0.10	0.25	
Efficacy rate	0.05	0.08	0.20	0.50	0.50	
Utility 2	0	0.1	0.1	0.4	0.2	
Utility 3	42.6	44	50	66	60	
MTD selection (%)						
U2 - shared alpha	0	0.1	0.6	6.6	92.7	0
U2 - dose-specific alpha	0	0.1	0.6	6.3	93	0
U3 - shared alpha	0	0.1	0.6	6.4	92.9	0
U3 - dose-specific alpha	0	0.1	0.6	6.6	92.7	0
OBD selection (%)						
U2 - shared alpha	0.3	1.7	3.9	46.9	47.1	0.1
U2 - dose-specific alpha	0.5	2.4	3.6	48.4	45	0.1
U3 - shared alpha	1.5	2.2	4.5	48.6	43.1	0.1
U3 - dose-specific alpha	1.3	2.9	5.1	51.6	39	0.1
Mean administrations						
U2 - shared alpha	3.1	3.1	3.5	8.3	11.9	-
U2 - dose-specific alpha	3.2	3.1	3.5	8.4	11.8	-
U3 - shared alpha	3.1	3.1	3.5	8.2	12	-
U3 - dose-specific alpha	3.1	3.2	3.5	8.4	11.8	-
Mean IPDE administrations						
U2 - shared alpha	0.1	2.4	1.1	3.2	3.3	-
U2 - dose-specific alpha	0.1	2.4	1.1	3.2	3.2	-
U3 - shared alpha	0.1	2.4	1.1	3.2	3.3	-
U3 - dose-specific alpha	0.1	2.4	1.1	3.2	3.2	-
Mean trial duration (days)						Days
U2 - shared alpha						1102.4
U2 - dose-specific alpha						1102.2
U3 - shared alpha						1102.6
U3 - dose-specific alpha						1102.5

Priors: efficacy beta-binomial Beta(0.5, 0.5); efficacy additive alpha Beta(0.15, 0.85); toxicity additive alpha Beta(0.15, 0.85).

Final column: early stopping % for MTD selection and no-OBD % for OBD selection; shaded rows are the dose-specific alpha efficacy model.

Phase I/II Operating Characteristics

Shared vs dose-specific additive-alpha efficacy - IPDE alpha = 0.9 - allocation = two-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No MTD/OBD %
Scenario 25						
DLT rate	0.09	0.18	0.37	0.63	0.77	
Efficacy rate	0.20	0.25	0.40	0.45	0.50	
Utility 2	0.1	0.1	0	-0.2	-0.3	
Utility 3	48.4	47.8	49.2	41.8	39.2	
MTD selection (%)						
U2 - shared alpha	16.4	44	36.2	3.2	0	0.2
U2 - dose-specific alpha	17	42.9	36.8	3.1	0	0.2
U3 - shared alpha	16.6	43.5	36.6	3.1	0	0.2
U3 - dose-specific alpha	16.7	43	37.3	2.8	0	0.2
OBD selection (%)						
U2 - shared alpha	37.1	37.3	22.1	1.8	0	1.7
U2 - dose-specific alpha	37	36.8	22.7	1.7	0	1.8
U3 - shared alpha	39.9	36.4	20.3	1.6	0	1.8
U3 - dose-specific alpha	40	37.2	19.6	1.5	0	1.7
Mean administrations						
U2 - shared alpha	10.7	9.3	8	1.6	0.2	-
U2 - dose-specific alpha	10.6	9.4	8	1.7	0.2	-
U3 - shared alpha	10.6	9.3	7.9	1.6	0.2	-
U3 - dose-specific alpha	10.6	9.4	7.9	1.6	0.2	-
Mean IPDE administrations						
U2 - shared alpha	3.1	4.1	2.2	0.6	0	-
U2 - dose-specific alpha	3	4.1	2.2	0.6	0	-
U3 - shared alpha	3.1	4.1	2.2	0.6	0	-
U3 - dose-specific alpha	3.1	4.1	2.2	0.6	0	-
Mean trial duration (days)						Days
U2 - shared alpha						1090.9
U2 - dose-specific alpha						1089.5
U3 - shared alpha						1090
U3 - dose-specific alpha						1089.4

Scenario 26						
DLT rate	0.15	0.22	0.30	0.62	0.80	
Efficacy rate	0.10	0.20	0.35	0.45	0.55	
Utility 2	-0.1	-0	0.1	-0.2	-0.2	
Utility 3	40	43.2	49	42.2	41	
MTD selection (%)						
U2 - shared alpha	37.2	36.1	22.4	3.4	0	0.9
U2 - dose-specific alpha	37.4	35.2	23.1	3.4	0	0.9
U3 - shared alpha	37.4	35.1	23.5	3.2	0	0.8
U3 - dose-specific alpha	37.3	35.7	23.1	3	0	0.9
OBD selection (%)						
U2 - shared alpha	31.5	29.1	17.7	1.9	0	19.8
U2 - dose-specific alpha	30.8	29.2	17.6	2.1	0	20.3
U3 - shared alpha	34.1	28	16.6	1.5	0	19.8
U3 - dose-specific alpha	34.1	28.9	15.5	1.6	0	19.9
Mean administrations						
U2 - shared alpha	11.8	7.2	6.3	1.5	0.1	-
U2 - dose-specific alpha	11.8	7.2	6.3	1.5	0.1	-
U3 - shared alpha	11.8	7.3	6.2	1.5	0.1	-
U3 - dose-specific alpha	11.8	7.3	6.2	1.5	0.1	-
Mean IPDE administrations						
U2 - shared alpha	3.6	3.4	1.9	0.6	0	-
U2 - dose-specific alpha	3.7	3.4	1.9	0.6	0	-
U3 - shared alpha	3.7	3.4	1.9	0.6	0	-
U3 - dose-specific alpha	3.7	3.5	1.9	0.6	0	-
Mean trial duration (days)						Days
U2 - shared alpha						951.4
U2 - dose-specific alpha						951.3
U3 - shared alpha						950.7
U3 - dose-specific alpha						950.8

Priors: efficacy beta-binomial Beta(0.5, 0.5); efficacy additive alpha Beta(0.15, 0.85); toxicity additive alpha Beta(0.15, 0.85).

Final column: early stopping % for MTD selection and no-OBD % for OBD selection; shaded rows are the dose-specific alpha efficacy model.

Phase I/II Operating Characteristics

Shared vs dose-specific additive-alpha efficacy - IPDE alpha = 0.9 - allocation = two-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No MTD/OBD %
Scenario 27						
DLT rate	0.11	0.17	0.30	0.50	0.62	
Efficacy rate	0.10	0.20	0.40	0.42	0.45	
Utility 2	-0	0	0.1	-0.1	-0.2	
Utility 3	41.6	45.2	52	45.2	42.2	
MTD selection (%)						
U2 - shared alpha	20.3	34.3	38.6	6.5	0	0.3
U2 - dose-specific alpha	20.6	33.9	39.1	6.1	0	0.3
U3 - shared alpha	20.5	34.3	38.6	6.3	0	0.3
U3 - dose-specific alpha	20.5	34.2	38.2	6.8	0	0.3
OBD selection (%)						
U2 - shared alpha	21.2	31	32.8	3.3	0	11.7
U2 - dose-specific alpha	22.3	30.5	32.1	3.2	0	11.9
U3 - shared alpha	24.6	30.1	30.4	3	0	11.9
U3 - dose-specific alpha	25.1	29.9	29.5	3.5	0	12
Mean administrations						
U2 - shared alpha	8.9	7.5	8.7	2.6	0.5	-
U2 - dose-specific alpha	8.9	7.5	8.5	2.6	0.5	-
U3 - shared alpha	8.9	7.5	8.7	2.6	0.5	-
U3 - dose-specific alpha	8.9	7.5	8.6	2.6	0.5	-
Mean IPDE administrations						
U2 - shared alpha	2.5	3.6	2.5	0.8	0.1	-
U2 - dose-specific alpha	2.5	3.7	2.5	0.8	0.1	-
U3 - shared alpha	2.5	3.6	2.5	0.8	0.1	-
U3 - dose-specific alpha	2.5	3.6	2.5	0.8	0.1	-
Mean trial duration (days)						Days
U2 - shared alpha						1019.3
U2 - dose-specific alpha						1018.6
U3 - shared alpha						1019.8
U3 - dose-specific alpha						1018.3

Scenario 28						
DLT rate	0.20	0.30	0.50	0.60	0.77	
Efficacy rate	0.05	0.30	0.32	0.35	0.40	
Utility 2	-0.1	0	-0.2	-0.2	-0.4	
Utility 3	35	46	39.2	37	33.2	
MTD selection (%)						
U2 - shared alpha	69.5	23.6	2.1	0	0	4.8
U2 - dose-specific alpha	69.9	23.2	2.1	0	0	4.8
U3 - shared alpha	69.5	23.6	2.2	0	0	4.7
U3 - dose-specific alpha	69.8	23.2	2.2	0	0	4.8
OBD selection (%)						
U2 - shared alpha	20	22.4	0.5	0	0	57.1
U2 - dose-specific alpha	20.3	21.7	0.6	0	0	57.4
U3 - shared alpha	20.1	21.7	0.5	0	0	57.7
U3 - dose-specific alpha	20.8	21.4	0.5	0	0	57.3
Mean administrations						
U2 - shared alpha	10.9	6.9	2.3	0.3	0	-
U2 - dose-specific alpha	10.9	6.9	2.3	0.3	0	-
U3 - shared alpha	10.9	6.9	2.3	0.3	0	-
U3 - dose-specific alpha	10.9	6.9	2.3	0.3	0	-
Mean IPDE administrations						
U2 - shared alpha	3.2	3	0.7	0.1	0	-
U2 - dose-specific alpha	3.2	3	0.7	0.1	0	-
U3 - shared alpha	3.2	3	0.7	0.1	0	-
U3 - dose-specific alpha	3.2	3	0.7	0.1	0	-
Mean trial duration (days)						Days
U2 - shared alpha						731.7
U2 - dose-specific alpha						730.5
U3 - shared alpha						730.6
U3 - dose-specific alpha						732.4

Priors: efficacy beta-binomial Beta(0.5, 0.5); efficacy additive alpha Beta(0.15, 0.85); toxicity additive alpha Beta(0.15, 0.85).

Final column: early stopping % for MTD selection and no-OBD % for OBD selection; shaded rows are the dose-specific alpha efficacy model.

Phase I/II Operating Characteristics

Shared vs dose-specific additive-alpha efficacy - IPDE alpha = 0.9 - allocation = two-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No MTD/OBD %
Scenario 29						
DLT rate	0.19	0.30	0.56	0.74	0.86	
Efficacy rate	0.40	0.40	0.40	0.40	0.40	
Utility 2	0.2	0.1	-0.2	-0.3	-0.5	
Utility 3	56.4	52	41.6	34.4	29.6	
MTD selection (%)						
U2 - shared alpha	58.6	33.5	4.5	0	0	3.4
U2 - dose-specific alpha	58.8	34	3.8	0	0	3.4
U3 - shared alpha	58.7	34.1	3.7	0	0	3.5
U3 - dose-specific alpha	59	32.9	4.7	0	0	3.4
OBD selection (%)						
U2 - shared alpha	75.7	19.2	1.4	0	0	3.7
U2 - dose-specific alpha	77	18.2	1	0	0	3.8
U3 - shared alpha	78.2	17.1	1	0	0	3.7
U3 - dose-specific alpha	78.2	16.8	1.2	0	0	3.8
Mean administrations						
U2 - shared alpha	20.1	6.7	2.4	0.2	0	-
U2 - dose-specific alpha	20.1	6.7	2.3	0.2	0	-
U3 - shared alpha	20.1	6.7	2.3	0.2	0	-
U3 - dose-specific alpha	20.1	6.6	2.3	0.2	0	-
Mean IPDE administrations						
U2 - shared alpha	5.3	2.3	0.7	0.1	0	-
U2 - dose-specific alpha	5.3	2.3	0.6	0.1	0	-
U3 - shared alpha	5.3	2.3	0.7	0.1	0	-
U3 - dose-specific alpha	5.3	2.3	0.7	0.1	0	-
Mean trial duration (days)						Days
U2 - shared alpha						1158.7
U2 - dose-specific alpha						1158.1
U3 - shared alpha						1158.9
U3 - dose-specific alpha						1158.1

Scenario 30						
DLT rate	0.05	0.08	0.17	0.27	0.54	
Efficacy rate	0.12	0.15	0.20	0.40	0.50	
Utility 2	0.1	0.1	0	0.1	-0	
Utility 3	45.2	45.8	45.2	53.2	48.4	
MTD selection (%)						
U2 - shared alpha	3.1	3.4	36	46.8	10.7	0
U2 - dose-specific alpha	3.1	3.6	35.5	47.9	9.9	0
U3 - shared alpha	3.1	3.6	35.9	47.4	10	0
U3 - dose-specific alpha	3.1	3.7	36.1	46.8	10.3	0
OBD selection (%)						
U2 - shared alpha	17.4	20.3	20.6	35.1	5	1.6
U2 - dose-specific alpha	18.7	19.3	21	35	4.4	1.6
U3 - shared alpha	24.5	21.4	17.7	31.5	3.3	1.6
U3 - dose-specific alpha	24.5	22.4	17.1	31	3.4	1.6
Mean administrations						
U2 - shared alpha	5.4	5.3	6.8	8	4.2	-
U2 - dose-specific alpha	5.4	5.4	6.8	8	4.2	-
U3 - shared alpha	5.4	5.3	6.8	8	4.2	-
U3 - dose-specific alpha	5.4	5.3	6.7	8.1	4.2	-
Mean IPDE administrations						
U2 - shared alpha	1.1	3.1	2.2	2.8	1.1	-
U2 - dose-specific alpha	1.1	3.1	2.2	2.8	1.1	-
U3 - shared alpha	1.1	3.1	2.2	2.8	1.1	-
U3 - dose-specific alpha	1.1	3.1	2.2	2.8	1.1	-
Mean trial duration (days)						Days
U2 - shared alpha						1073.2
U2 - dose-specific alpha						1072.4
U3 - shared alpha						1072.2
U3 - dose-specific alpha						1072.7

Priors: efficacy beta-binomial Beta(0.5, 0.5); efficacy additive alpha Beta(0.15, 0.85); toxicity additive alpha Beta(0.15, 0.85).

Final column: early stopping % for MTD selection and no-OBD % for OBD selection; shaded rows are the dose-specific alpha efficacy model.

Phase I/II Operating Characteristics

Shared vs dose-specific additive-alpha efficacy - IPDE alpha = 0.9 - allocation = two-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No MTD/OBD %
Scenario 31						
DLT rate	0.04	0.06	0.10	0.20	0.60	
Efficacy rate	0.20	0.50	0.50	0.50	0.50	
Utility 2	0.2	0.4	0.4	0.3	-0.1	
Utility 3	50.4	67.6	66	62	46	
MTD selection (%)						
U2 - shared alpha	2.2	1.6	12.5	50.9	32.8	0
U2 - dose-specific alpha	2.2	1.5	12.2	51.2	32.9	0
U3 - shared alpha	2.2	1.5	12.9	49.7	33.7	0
U3 - dose-specific alpha	2.2	1.6	13	50.8	32.4	0
OBD selection (%)						
U2 - shared alpha	4.5	34.5	30.5	25.6	4.8	0.1
U2 - dose-specific alpha	5.3	37.6	29.3	23.3	4.4	0.1
U3 - shared alpha	6.1	41.2	28.7	20.9	3	0.1
U3 - dose-specific alpha	7.2	46	25.2	18.9	2.6	0.1
Mean administrations						
U2 - shared alpha	4.2	5.9	6.7	8.2	5.1	-
U2 - dose-specific alpha	4.2	5.9	6.7	8.1	5.1	-
U3 - shared alpha	4.2	5.8	6.8	8.1	5.1	-
U3 - dose-specific alpha	4.2	5.9	6.7	8.2	5.1	-
Mean IPDE administrations						
U2 - shared alpha	0.5	2.9	1.7	2.1	1.4	-
U2 - dose-specific alpha	0.5	2.9	1.7	2.1	1.4	-
U3 - shared alpha	0.5	2.9	1.7	2.1	1.4	-
U3 - dose-specific alpha	0.5	2.9	1.7	2.1	1.4	-
Mean trial duration (days)						Days
U2 - shared alpha						1180.7
U2 - dose-specific alpha						1181.4
U3 - shared alpha						1180.9
U3 - dose-specific alpha						1181.7

Scenario 32						
DLT rate	0.02	0.05	0.11	0.30	0.36	
Efficacy rate	0.08	0.15	0.30	0.30	0.30	
Utility 2	0.1	0.1	0.2	0	-0.1	
Utility 3	44	47	53.6	46	43.6	
MTD selection (%)						
U2 - shared alpha	0.8	1	13.5	48.7	36	0
U2 - dose-specific alpha	0.8	0.9	14.3	48.5	35.5	0
U3 - shared alpha	0.8	1.3	13.2	48.9	35.8	0
U3 - dose-specific alpha	0.8	1.1	13.7	49.5	34.9	0
OBD selection (%)						
U2 - shared alpha	9.4	13.5	35.1	31.5	9.7	0.8
U2 - dose-specific alpha	9.5	13.9	34.2	33	8.7	0.7
U3 - shared alpha	13.1	19.3	34.2	26.9	5.8	0.7
U3 - dose-specific alpha	14.4	19	33.6	27.4	4.9	0.7
Mean administrations						
U2 - shared alpha	3.8	4.5	7.8	7.9	5.9	-
U2 - dose-specific alpha	3.8	4.5	7.8	8	5.8	-
U3 - shared alpha	3.8	4.5	7.9	7.9	5.9	-
U3 - dose-specific alpha	3.8	4.5	7.8	8	5.8	-
Mean IPDE administrations						
U2 - shared alpha	0.4	2.9	2.5	2.9	2	-
U2 - dose-specific alpha	0.4	2.9	2.5	3	2	-
U3 - shared alpha	0.3	2.9	2.5	2.9	2	-
U3 - dose-specific alpha	0.3	2.9	2.5	2.9	2	-
Mean trial duration (days)						Days
U2 - shared alpha						1055.8
U2 - dose-specific alpha						1055.8
U3 - shared alpha						1057.4
U3 - dose-specific alpha						1056.3

Priors: efficacy beta-binomial Beta(0.5, 0.5); efficacy additive alpha Beta(0.15, 0.85); toxicity additive alpha Beta(0.15, 0.85).

Final column: early stopping % for MTD selection and no-OBD % for OBD selection; shaded rows are the dose-specific alpha efficacy model.

Phase I/II Operating Characteristics

Shared vs dose-specific additive-alpha efficacy - IPDE alpha = 0.9 - allocation = two-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No MTD/OBD %
Scenario 33						
DLT rate	0.01	0.02	0.09	0.32	0.38	
Efficacy rate	0.05	0.08	0.20	0.40	0.50	
Utility 2	0	0.1	0.1	0.1	0.1	
Utility 3	42.6	44	48.4	51.2	54.8	
MTD selection (%)						
U2 - shared alpha	0	1.1	17.6	53.1	28.2	0
U2 - dose-specific alpha	0	0.8	17	53.9	28.3	0
U3 - shared alpha	0	1	17.2	53.8	28	0
U3 - dose-specific alpha	0	0.8	16.6	54.4	28.2	0
OBD selection (%)						
U2 - shared alpha	8.6	9.7	18.5	45.3	17.8	0.1
U2 - dose-specific alpha	9.6	9.7	16.9	46.1	17.6	0.1
U3 - shared alpha	13.7	11.7	15.8	43.2	15.5	0.1
U3 - dose-specific alpha	13.2	11.6	16.2	43.9	14.9	0.2
Mean administrations						
U2 - shared alpha	3.5	3.7	5.5	8.4	8.9	-
U2 - dose-specific alpha	3.5	3.7	5.6	8.4	8.8	-
U3 - shared alpha	3.4	3.7	5.6	8.4	8.8	-
U3 - dose-specific alpha	3.5	3.7	5.6	8.4	8.8	-
Mean IPDE administrations						
U2 - shared alpha	0.2	2.7	1.7	3.1	2.2	-
U2 - dose-specific alpha	0.2	2.7	1.7	3.1	2.3	-
U3 - shared alpha	0.2	2.7	1.8	3.1	2.2	-
U3 - dose-specific alpha	0.2	2.7	1.7	3.1	2.2	-
Mean trial duration (days)						Days
U2 - shared alpha						1101
U2 - dose-specific alpha						1100.2
U3 - shared alpha						1100.8
U3 - dose-specific alpha						1100.7

Scenario 34						
DLT rate	0.05	0.13	0.30	0.36	0.42	
Efficacy rate	0.05	0.10	0.30	0.32	0.35	
Utility 2	0	-0	0	-0	-0.1	
Utility 3	41	40.8	46	44.8	44.2	
MTD selection (%)						
U2 - shared alpha	4.8	21.6	50.4	19	4.2	0
U2 - dose-specific alpha	4.5	21.8	51.6	17.8	4.3	0
U3 - shared alpha	5	21.5	50.9	18.2	4.4	0
U3 - dose-specific alpha	5.1	21.3	51.7	17.4	4.5	0
OBD selection (%)						
U2 - shared alpha	15.4	20.3	44.6	12	1.7	6
U2 - dose-specific alpha	16	20	45.7	10.8	1.5	6
U3 - shared alpha	21	21	41.1	9.7	1	6.2
U3 - dose-specific alpha	21.1	19.6	42.7	9	1.3	6.3
Mean administrations						
U2 - shared alpha	5.5	6.7	10.3	4.6	1.9	-
U2 - dose-specific alpha	5.5	6.8	10.3	4.5	1.9	-
U3 - shared alpha	5.5	6.7	10.3	4.5	1.9	-
U3 - dose-specific alpha	5.5	6.7	10.4	4.5	1.9	-
Mean IPDE administrations						
U2 - shared alpha	1.2	3.6	3.3	1.6	0.6	-
U2 - dose-specific alpha	1.1	3.7	3.3	1.6	0.6	-
U3 - shared alpha	1.2	3.6	3.3	1.6	0.6	-
U3 - dose-specific alpha	1.1	3.6	3.3	1.6	0.6	-
Mean trial duration (days)						Days
U2 - shared alpha						1022.8
U2 - dose-specific alpha						1021.2
U3 - shared alpha						1022.9
U3 - dose-specific alpha						1022.4

Priors: efficacy beta-binomial Beta(0.5, 0.5); efficacy additive alpha Beta(0.15, 0.85); toxicity additive alpha Beta(0.15, 0.85).

Final column: early stopping % for MTD selection and no-OBD % for OBD selection; shaded rows are the dose-specific alpha efficacy model.

Phase I/II Operating Characteristics

Shared vs dose-specific additive-alpha efficacy - IPDE alpha = 0.9 - allocation = two-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No MTD/OBD %
Scenario 35						
DLT rate	0.03	0.09	0.30	0.40	0.45	
Efficacy rate	0.20	0.45	0.45	0.45	0.45	
Utility 2	0.2	0.4	0.1	0.1	0	
Utility 3	50.8	63.4	55	51	49	
MTD selection (%)						
U2 - shared alpha	1.9	13.7	54.6	25.1	4.7	0
U2 - dose-specific alpha	1.9	13.5	53.6	25.2	5.8	0
U3 - shared alpha	1.9	12.8	54.4	25.8	5.1	0
U3 - dose-specific alpha	1.9	13.6	53.5	25.5	5.5	0
OBD selection (%)						
U2 - shared alpha	10.2	47.3	32.2	9.4	0.8	0.1
U2 - dose-specific alpha	10.1	49.5	29.8	9.6	0.9	0.1
U3 - shared alpha	12.1	50.4	29.7	7.2	0.5	0.1
U3 - dose-specific alpha	12.1	53	26.5	7.5	0.8	0.1
Mean administrations						
U2 - shared alpha	4.6	9.6	9.5	4.5	1.8	-
U2 - dose-specific alpha	4.6	9.7	9.4	4.5	1.8	-
U3 - shared alpha	4.7	9.6	9.5	4.4	1.8	-
U3 - dose-specific alpha	4.6	9.6	9.5	4.5	1.8	-
Mean IPDE administrations						
U2 - shared alpha	0.7	4	2.4	1.4	0.5	-
U2 - dose-specific alpha	0.7	4.1	2.3	1.4	0.5	-
U3 - shared alpha	0.7	4	2.3	1.4	0.5	-
U3 - dose-specific alpha	0.7	4	2.3	1.4	0.5	-
Mean trial duration (days)						Days
U2 - shared alpha						1161.8
U2 - dose-specific alpha						1161
U3 - shared alpha						1160.7
U3 - dose-specific alpha						1160.8

Scenario 36						
DLT rate	0.13	0.29	0.35	0.40	0.43	
Efficacy rate	0.05	0.20	0.22	0.30	0.35	
Utility 2	-0.1	-0.1	-0.1	-0.1	-0.1	
Utility 3	37.8	40.4	39.2	42	43.8	
MTD selection (%)						
U2 - shared alpha	42.8	38.5	14.8	2.3	0.7	0.9
U2 - dose-specific alpha	42.3	39.3	14.4	2.4	0.7	0.9
U3 - shared alpha	42.8	38.3	15	2.5	0.5	0.9
U3 - dose-specific alpha	43.3	38.4	14.2	2.5	0.7	0.9
OBD selection (%)						
U2 - shared alpha	21.6	35.5	7.9	1.3	0.5	33.2
U2 - dose-specific alpha	20.5	36.9	7.9	1.5	0.4	32.8
U3 - shared alpha	24	34.9	7	1.3	0.3	32.5
U3 - dose-specific alpha	24.8	33.7	6.8	1.1	0.4	33.2
Mean administrations						
U2 - shared alpha	9.6	8.4	4.5	1.3	0.4	-
U2 - dose-specific alpha	9.5	8.5	4.5	1.4	0.4	-
U3 - shared alpha	9.6	8.4	4.6	1.4	0.4	-
U3 - dose-specific alpha	9.6	8.4	4.5	1.4	0.4	-
Mean IPDE administrations						
U2 - shared alpha	2.7	4	1.5	0.6	0.1	-
U2 - dose-specific alpha	2.7	4	1.5	0.6	0.1	-
U3 - shared alpha	2.7	4	1.5	0.6	0.1	-
U3 - dose-specific alpha	2.7	4	1.5	0.6	0.1	-
Mean trial duration (days)						Days
U2 - shared alpha						844.4
U2 - dose-specific alpha						846.3
U3 - shared alpha						846.7
U3 - dose-specific alpha						845.5

Priors: efficacy beta-binomial Beta(0.5, 0.5); efficacy additive alpha Beta(0.15, 0.85); toxicity additive alpha Beta(0.15, 0.85).

Final column: early stopping % for MTD selection and no-OBD % for OBD selection; shaded rows are the dose-specific alpha efficacy model.

Phase I/II Operating Characteristics

Shared vs dose-specific additive-alpha efficacy - IPDE alpha = 0.9 - allocation = two-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No MTD/OBD %
Scenario 37						
DLT rate	0.08	0.30	0.40	0.45	0.50	
Efficacy rate	0.05	0.20	0.20	0.20	0.20	
Utility 2	-0	-0.1	-0.2	-0.2	-0.3	
Utility 3	39.8	40	36	34	32	
MTD selection (%)						
U2 - shared alpha	37.2	46.5	15.3	0.8	0	0.2
U2 - dose-specific alpha	37.4	46.7	15.1	0.5	0.1	0.2
U3 - shared alpha	36.9	47.5	14.6	0.8	0	0.2
U3 - dose-specific alpha	37.1	47.5	14.6	0.5	0.1	0.2
OBD selection (%)						
U2 - shared alpha	27.5	43.5	7.4	0.2	0	21.4
U2 - dose-specific alpha	28.8	42.5	6.8	0	0.1	21.8
U3 - shared alpha	30.5	42.3	5.2	0.2	0	21.8
U3 - dose-specific alpha	30.1	42.6	5.3	0	0	22
Mean administrations						
U2 - shared alpha	8.9	11	5	1.2	0.2	-
U2 - dose-specific alpha	8.9	10.9	5	1.1	0.2	-
U3 - shared alpha	8.9	11	5	1.2	0.2	-
U3 - dose-specific alpha	8.9	11	4.9	1.1	0.2	-
Mean IPDE administrations						
U2 - shared alpha	2.4	5.1	1.6	0.6	0.1	-
U2 - dose-specific alpha	2.4	5.1	1.6	0.5	0.1	-
U3 - shared alpha	2.4	5.1	1.6	0.6	0.1	-
U3 - dose-specific alpha	2.4	5.1	1.6	0.5	0.1	-
Mean trial duration (days)						Days
U2 - shared alpha						907.5
U2 - dose-specific alpha						908.8
U3 - shared alpha						908.9
U3 - dose-specific alpha						907.2

Priors: efficacy beta-binomial Beta(0.5, 0.5); efficacy additive alpha Beta(0.15, 0.85); toxicity additive alpha Beta(0.15, 0.85).

Final column: early stopping % for MTD selection and no-OBD % for OBD selection; shaded rows are the dose-specific alpha efficacy model.

Phase I/II Operating Characteristics

Shared vs dose-specific additive-alpha efficacy - IPDE alpha = 0.9 - allocation = one-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No MTD/OBD %
Scenario 1						
DLT rate	0.30	0.35	0.40	0.45	0.50	
Efficacy rate	0.05	0.06	0.07	0.08	0.09	
Utility 2	-0.2	-0.3	-0.3	-0.4	-0.4	
Utility 3	31	29.6	28.2	26.8	25.4	
MTD selection (%)						
U2 - shared alpha	70.8	7.5	1.6	0	0	20.1
U2 - dose-specific alpha	71.1	7.3	1.5	0	0	20.1
U3 - shared alpha	71.4	7.3	1.3	0.1	0	19.9
U3 - dose-specific alpha	71.1	7.3	1.5	0	0	20.1
OBD selection (%)						
U2 - shared alpha	12	3.2	0.3	0	0	84.5
U2 - dose-specific alpha	12.3	2.4	0.3	0	0	85
U3 - shared alpha	11.8	2.7	0.3	0	0	85.2
U3 - dose-specific alpha	12.3	2.5	0.4	0	0	84.8
Mean administrations						
U2 - shared alpha	11.6	4.3	1	0.4	0	-
U2 - dose-specific alpha	11.6	4.3	1	0.4	0	-
U3 - shared alpha	11.7	4.2	1	0.4	0	-
U3 - dose-specific alpha	11.7	4.2	1	0.4	0.1	-
Mean IPDE administrations						
U2 - shared alpha	3.2	2.4	0.3	0.2	0	-
U2 - dose-specific alpha	3.2	2.4	0.4	0.2	0	-
U3 - shared alpha	3.3	2.4	0.3	0.2	0	-
U3 - dose-specific alpha	3.3	2.4	0.4	0.2	0	-
Mean trial duration (days)						Days
U2 - shared alpha						607.2
U2 - dose-specific alpha						606.4
U3 - shared alpha						606.2
U3 - dose-specific alpha						607.8

Scenario 2						
DLT rate	0.15	0.20	0.30	0.35	0.45	
Efficacy rate	0.20	0.25	0.50	0.50	0.50	
Utility 2	0.1	0.1	0.2	0.1	0.1	
Utility 3	46	47	58	56	52	
MTD selection (%)						
U2 - shared alpha	15	40.5	36.6	7.3	0.3	0.3
U2 - dose-specific alpha	14.9	43	35.3	6.2	0.3	0.3
U3 - shared alpha	14.8	43.4	35	6.3	0.2	0.3
U3 - dose-specific alpha	14.4	46	33.5	5.6	0.1	0.4
OBD selection (%)						
U2 - shared alpha	32.9	25.9	32.6	5.9	0.2	2.5
U2 - dose-specific alpha	33.8	27.2	31.2	4.8	0.3	2.7
U3 - shared alpha	35.3	27.5	29.9	4.8	0.1	2.4
U3 - dose-specific alpha	38.1	27.2	28	4	0.1	2.6
Mean administrations						
U2 - shared alpha	11.5	8.9	7.5	1.7	0.1	-
U2 - dose-specific alpha	11.6	8.9	7.3	1.7	0.1	-
U3 - shared alpha	12.1	9	7.1	1.4	0	-
U3 - dose-specific alpha	12.2	9.1	6.9	1.4	0	-
Mean IPDE administrations						
U2 - shared alpha	3	3.9	2.2	0.6	0	-
U2 - dose-specific alpha	3.1	3.9	2.2	0.6	0	-
U3 - shared alpha	3.3	4	2.1	0.5	0	-
U3 - dose-specific alpha	3.3	4	2.1	0.5	0	-
Mean trial duration (days)						Days
U2 - shared alpha						1093.5
U2 - dose-specific alpha						1094.7
U3 - shared alpha						1090.5
U3 - dose-specific alpha						1088.3

Priors: efficacy beta-binomial Beta(0.5, 0.5); efficacy additive alpha Beta(0.15, 0.85); toxicity additive alpha Beta(0.15, 0.85).

Final column: early stopping % for MTD selection and no-OBD % for OBD selection; shaded rows are the dose-specific alpha efficacy model.

Phase I/II Operating Characteristics

Shared vs dose-specific additive-alpha efficacy - IPDE alpha = 0.9 - allocation = one-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No MTD/OBD %
Scenario 3						
DLT rate	0.15	0.20	0.30	0.35	0.45	
Efficacy rate	0.10	0.40	0.40	0.40	0.40	
Utility 2	-0.1	0.2	0.1	0.1	-0.1	
Utility 3	40	56	52	50	46	
MTD selection (%)						
U2 - shared alpha	17.8	46	29.3	6.2	0.4	0.3
U2 - dose-specific alpha	17.7	46.1	29.2	6.6	0.1	0.3
U3 - shared alpha	18.2	49.3	26.8	5.3	0.1	0.3
U3 - dose-specific alpha	17.7	50.6	25.8	5.5	0.1	0.3
OBD selection (%)						
U2 - shared alpha	12.1	53.7	18.8	4	0.2	11.2
U2 - dose-specific alpha	12.1	53.5	19.2	3.9	0.1	11.2
U3 - shared alpha	13.2	56.7	16.3	3	0	10.8
U3 - dose-specific alpha	13	56.9	15.8	3.1	0	11.2
Mean administrations						
U2 - shared alpha	8.5	12.9	5.3	1.7	0.1	-
U2 - dose-specific alpha	8.4	13	5.3	1.7	0.1	-
U3 - shared alpha	8.7	13.6	4.8	1.5	0.1	-
U3 - dose-specific alpha	8.6	13.6	4.7	1.5	0.1	-
Mean IPDE administrations						
U2 - shared alpha	1.9	5.3	1.6	0.7	0.1	-
U2 - dose-specific alpha	1.9	5.3	1.6	0.7	0.1	-
U3 - shared alpha	2	5.6	1.4	0.6	0	-
U3 - dose-specific alpha	1.9	5.6	1.4	0.6	0	-
Mean trial duration (days)						Days
U2 - shared alpha						1043.4
U2 - dose-specific alpha						1044.1
U3 - shared alpha						1042
U3 - dose-specific alpha						1040.7

Scenario 4						
DLT rate	0.05	0.10	0.18	0.30	0.40	
Efficacy rate	0.05	0.10	0.25	0.50	0.55	
Utility 2	0	0	0.1	0.2	0.1	
Utility 3	41	42	47.8	58	57	
MTD selection (%)						
U2 - shared alpha	1.3	5.1	29.7	56.9	7	0
U2 - dose-specific alpha	1.1	4.6	30.6	56.8	6.9	0
U3 - shared alpha	1.2	4.9	30.6	57.3	6	0
U3 - dose-specific alpha	1.2	5.3	29.8	58.1	5.6	0
OBD selection (%)						
U2 - shared alpha	6.2	9.5	25.3	51.4	5.5	2.1
U2 - dose-specific alpha	6.6	9.9	26.3	50	5.4	1.8
U3 - shared alpha	8.2	10.8	24.6	50.3	4.1	2
U3 - dose-specific alpha	8.8	11.4	23.6	50.3	3.9	2
Mean administrations						
U2 - shared alpha	4.5	5.3	8.1	10.2	1.7	-
U2 - dose-specific alpha	4.4	5.3	8.2	10.2	1.7	-
U3 - shared alpha	4.7	5.5	8.1	10	1.4	-
U3 - dose-specific alpha	4.7	5.5	8.2	10	1.4	-
Mean IPDE administrations						
U2 - shared alpha	0.6	3.3	2.4	3.3	0.6	-
U2 - dose-specific alpha	0.6	3.3	2.4	3.3	0.6	-
U3 - shared alpha	0.7	3.4	2.5	3.3	0.5	-
U3 - dose-specific alpha	0.7	3.4	2.5	3.3	0.5	-
Mean trial duration (days)						Days
U2 - shared alpha						1074.5
U2 - dose-specific alpha						1075.4
U3 - shared alpha						1070.9
U3 - dose-specific alpha						1070.1

Priors: efficacy beta-binomial Beta(0.5, 0.5); efficacy additive alpha Beta(0.15, 0.85); toxicity additive alpha Beta(0.15, 0.85).

Final column: early stopping % for MTD selection and no-OBD % for OBD selection; shaded rows are the dose-specific alpha efficacy model.

Phase I/II Operating Characteristics

Shared vs dose-specific additive-alpha efficacy - IPDE alpha = 0.9 - allocation = one-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No MTD/OBD %
Scenario 5						
DLT rate	0.07	0.12	0.17	0.22	0.30	
Efficacy rate	0.05	0.30	0.30	0.30	0.30	
Utility 2	-0	0.2	0.1	0.1	0	
Utility 3	40.2	53.2	51.2	49.2	46	
MTD selection (%)						
U2 - shared alpha	2.2	8.2	25.2	40.5	23.9	0
U2 - dose-specific alpha	2.2	9.2	24.3	40.7	23.6	0
U3 - shared alpha	2.2	11.1	27.8	39.1	19.8	0
U3 - dose-specific alpha	2.2	10.9	28.2	38.1	20.6	0
OBD selection (%)						
U2 - shared alpha	4.1	31.7	29.4	24.5	8.1	2.2
U2 - dose-specific alpha	3.9	33.8	27.8	24.9	7.5	2.1
U3 - shared alpha	6	40.9	27.2	18.8	5	2.1
U3 - dose-specific alpha	6.1	42.4	26.8	17.8	4.7	2.2
Mean administrations						
U2 - shared alpha	4.2	8.5	7.6	6.6	2.7	-
U2 - dose-specific alpha	4.2	8.5	7.7	6.6	2.7	-
U3 - shared alpha	4.4	9.4	7.6	6.2	2.1	-
U3 - dose-specific alpha	4.3	9.4	7.7	6.1	2.2	-
Mean IPDE administrations						
U2 - shared alpha	0.4	4.3	2.5	2.5	1	-
U2 - dose-specific alpha	0.4	4.3	2.6	2.5	1	-
U3 - shared alpha	0.5	4.6	2.5	2.4	0.8	-
U3 - dose-specific alpha	0.5	4.6	2.6	2.3	0.8	-
Mean trial duration (days)						Days
U2 - shared alpha						1039.2
U2 - dose-specific alpha						1038.5
U3 - shared alpha						1037.1
U3 - dose-specific alpha						1036.2

Scenario 6						
DLT rate	0.29	0.34	0.38	0.43	0.52	
Efficacy rate	0.30	0.32	0.35	0.38	0.40	
Utility 2	0	-0	-0	-0.1	-0.1	
Utility 3	46.4	45.6	45.8	45.6	43.2	
MTD selection (%)						
U2 - shared alpha	63.4	11.9	2	0.3	0	22.4
U2 - dose-specific alpha	62.7	12.7	2	0.2	0	22.4
U3 - shared alpha	62.6	12.8	1.4	0.3	0	22.9
U3 - dose-specific alpha	63	12.3	1.7	0.2	0	22.8
OBD selection (%)						
U2 - shared alpha	63.8	8.7	1	0.3	0	26.2
U2 - dose-specific alpha	63.4	9.2	1.1	0.2	0	26.1
U3 - shared alpha	64.3	8.1	0.5	0.3	0	26.8
U3 - dose-specific alpha	64.7	8.2	0.6	0.2	0	26.3
Mean administrations						
U2 - shared alpha	19.8	5	1	0.2	0	-
U2 - dose-specific alpha	19.7	5.1	1	0.2	0	-
U3 - shared alpha	20	4.9	0.8	0.2	0	-
U3 - dose-specific alpha	20.1	4.9	0.8	0.2	0	-
Mean IPDE administrations						
U2 - shared alpha	5.2	2.1	0.3	0.1	0	-
U2 - dose-specific alpha	5.2	2.2	0.3	0.1	0	-
U3 - shared alpha	5.3	2.1	0.3	0.1	0	-
U3 - dose-specific alpha	5.3	2.1	0.3	0.1	0	-
Mean trial duration (days)						Days
U2 - shared alpha						1006.2
U2 - dose-specific alpha						1003.1
U3 - shared alpha						998.6
U3 - dose-specific alpha						1003.6

Priors: efficacy beta-binomial Beta(0.5, 0.5); efficacy additive alpha Beta(0.15, 0.85); toxicity additive alpha Beta(0.15, 0.85).

Final column: early stopping % for MTD selection and no-OBD % for OBD selection; shaded rows are the dose-specific alpha efficacy model.

Phase I/II Operating Characteristics

Shared vs dose-specific additive-alpha efficacy - IPDE alpha = 0.9 - allocation = one-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No MTD/OBD %
Scenario 7						
DLT rate	0.16	0.20	0.26	0.31	0.36	
Efficacy rate	0.20	0.50	0.50	0.50	0.50	
Utility 2	0	0.3	0.2	0.2	0.1	
Utility 3	45.6	62	59.6	57.6	55.6	
MTD selection (%)						
U2 - shared alpha	15.7	62.6	16.9	3.8	0.5	0.5
U2 - dose-specific alpha	15.7	62.1	17.6	3.5	0.5	0.6
U3 - shared alpha	16.2	65.5	14.4	3.1	0.2	0.6
U3 - dose-specific alpha	16	65.5	15	2.9	0.1	0.5
OBD selection (%)						
U2 - shared alpha	17.4	62.1	14.8	2.5	0.4	2.8
U2 - dose-specific alpha	18.4	61.7	14.4	2.3	0.4	2.8
U3 - shared alpha	20	63.5	11.8	1.9	0.2	2.6
U3 - dose-specific alpha	19.1	64.5	12.2	1.5	0.1	2.6
Mean administrations						
U2 - shared alpha	9.8	15.4	3.6	0.7	0.1	-
U2 - dose-specific alpha	9.8	15.4	3.6	0.7	0.1	-
U3 - shared alpha	10.2	15.8	3	0.6	0	-
U3 - dose-specific alpha	10.2	15.8	3.1	0.5	0	-
Mean IPDE administrations						
U2 - shared alpha	2.2	5.6	1.1	0.2	0	-
U2 - dose-specific alpha	2.2	5.6	1	0.3	0	-
U3 - shared alpha	2.4	5.7	0.9	0.2	0	-
U3 - dose-specific alpha	2.4	5.7	0.9	0.2	0	-
Mean trial duration (days)						Days
U2 - shared alpha						1130.8
U2 - dose-specific alpha						1131.3
U3 - shared alpha						1129.2
U3 - dose-specific alpha						1128.8

Scenario 8						
DLT rate	0.32	0.37	0.42	0.48	0.52	
Efficacy rate	0.10	0.15	0.20	0.30	0.40	
Utility 2	-0.2	-0.2	-0.2	-0.2	-0.1	
Utility 3	33.2	34.2	35.2	38.8	43.2	
MTD selection (%)						
U2 - shared alpha	63.8	4.6	0.8	0	0	30.8
U2 - dose-specific alpha	63.9	4.7	0.7	0	0	30.7
U3 - shared alpha	63.4	4.8	0.7	0	0	31.1
U3 - dose-specific alpha	63.7	4.3	0.9	0	0	31.1
OBD selection (%)						
U2 - shared alpha	26.6	3.4	0.3	0	0	69.7
U2 - dose-specific alpha	27.2	3.2	0.3	0	0	69.3
U3 - shared alpha	27.2	2.8	0.3	0	0	69.7
U3 - dose-specific alpha	27.6	2.6	0.2	0	0	69.6
Mean administrations						
U2 - shared alpha	14.3	4.3	0.8	0.2	0	-
U2 - dose-specific alpha	14.4	4.3	0.8	0.2	0	-
U3 - shared alpha	14.5	4.2	0.8	0.2	0	-
U3 - dose-specific alpha	14.5	4.3	0.8	0.2	0	-
Mean IPDE administrations						
U2 - shared alpha	4.1	2.2	0.3	0.1	0	-
U2 - dose-specific alpha	4.1	2.3	0.3	0.1	0	-
U3 - shared alpha	4.1	2.2	0.3	0.1	0	-
U3 - dose-specific alpha	4.1	2.2	0.3	0.1	0	-
Mean trial duration (days)						Days
U2 - shared alpha						709.2
U2 - dose-specific alpha						712.9
U3 - shared alpha						711.6
U3 - dose-specific alpha						709.6

Priors: efficacy beta-binomial Beta(0.5, 0.5); efficacy additive alpha Beta(0.15, 0.85); toxicity additive alpha Beta(0.15, 0.85).

Final column: early stopping % for MTD selection and no-OBD % for OBD selection; shaded rows are the dose-specific alpha efficacy model.

Phase I/II Operating Characteristics

Shared vs dose-specific additive-alpha efficacy - IPDE alpha = 0.9 - allocation = one-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No MTD/OBD %
Scenario 9						
DLT rate	0.15	0.18	0.22	0.26	0.29	
Efficacy rate	0.05	0.08	0.20	0.25	0.50	
Utility 2	-0.1	-0.1	-0	-0	0.2	
Utility 3	37	37.6	43.2	44.6	58.4	
MTD selection (%)						
U2 - shared alpha	19.3	20.6	25.1	21.9	12.7	0.4
U2 - dose-specific alpha	19.5	20.7	24.3	22.6	12.5	0.4
U3 - shared alpha	19.8	21	24.9	22.1	11.8	0.4
U3 - dose-specific alpha	19.7	21.9	24	21.9	12.1	0.4
OBD selection (%)						
U2 - shared alpha	16.8	16.1	22.1	11.9	10.6	22.5
U2 - dose-specific alpha	17.4	15.4	21.7	11.9	10.6	23
U3 - shared alpha	19.6	17.1	20.3	11.4	8.6	23
U3 - dose-specific alpha	19.9	16.9	19.5	11.2	9.2	23.3
Mean administrations						
U2 - shared alpha	8.1	6.6	5.9	4.2	2.2	-
U2 - dose-specific alpha	8.1	6.6	5.9	4.2	2.3	-
U3 - shared alpha	8.4	6.6	5.8	4.2	2.1	-
U3 - dose-specific alpha	8.4	6.6	5.8	4.2	2.1	-
Mean IPDE administrations						
U2 - shared alpha	2	3.7	2.1	1.7	0.7	-
U2 - dose-specific alpha	2	3.7	2.1	1.7	0.8	-
U3 - shared alpha	2.1	3.7	2	1.7	0.7	-
U3 - dose-specific alpha	2.1	3.7	2	1.7	0.7	-
Mean trial duration (days)						Days
U2 - shared alpha						930.3
U2 - dose-specific alpha						929.8
U3 - shared alpha						928.2
U3 - dose-specific alpha						927.8

Scenario 10						
DLT rate	0.23	0.28	0.32	0.39	0.43	
Efficacy rate	0.05	0.40	0.42	0.45	0.50	
Utility 2	-0.2	0.1	0.1	0.1	0.1	
Utility 3	33.8	52.8	52.4	51.4	52.8	
MTD selection (%)						
U2 - shared alpha	60.8	20.8	10.8	1.9	0.1	5.6
U2 - dose-specific alpha	61.3	20.4	11	1.6	0.1	5.6
U3 - shared alpha	61.4	21.7	9.7	1.6	0	5.6
U3 - dose-specific alpha	61.2	21.9	9.6	1.6	0.1	5.6
OBD selection (%)						
U2 - shared alpha	14.2	23.3	7.8	1.3	0.1	53.3
U2 - dose-specific alpha	14.8	23.1	7.6	1.2	0.1	53.2
U3 - shared alpha	14.5	23.8	6.8	1.3	0	53.6
U3 - dose-specific alpha	14	24.2	6.9	1.1	0.1	53.7
Mean administrations						
U2 - shared alpha	10.7	8.6	2.6	0.6	0	-
U2 - dose-specific alpha	10.7	8.6	2.6	0.7	0	-
U3 - shared alpha	10.8	8.7	2.4	0.6	0	-
U3 - dose-specific alpha	10.8	8.8	2.4	0.6	0	-
Mean IPDE administrations						
U2 - shared alpha	2.7	3.9	0.8	0.3	0	-
U2 - dose-specific alpha	2.7	3.9	0.8	0.3	0	-
U3 - shared alpha	2.7	4	0.7	0.2	0	-
U3 - dose-specific alpha	2.7	4	0.7	0.2	0	-
Mean trial duration (days)						Days
U2 - shared alpha						820.8
U2 - dose-specific alpha						819.9
U3 - shared alpha						819
U3 - dose-specific alpha						819.9

Priors: efficacy beta-binomial Beta(0.5, 0.5); efficacy additive alpha Beta(0.15, 0.85); toxicity additive alpha Beta(0.15, 0.85).

Final column: early stopping % for MTD selection and no-OBD % for OBD selection; shaded rows are the dose-specific alpha efficacy model.

Phase I/II Operating Characteristics

Shared vs dose-specific additive-alpha efficacy - IPDE alpha = 0.9 - allocation = one-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No MTD/OBD %
Scenario 11						
DLT rate	0.22	0.32	0.44	0.59	0.67	
Efficacy rate	0.20	0.20	0.20	0.20	0.20	
Utility 2	-0	-0.1	-0.2	-0.4	-0.5	
Utility 3	43.2	39.2	34.4	28.4	25.2	
MTD selection (%)						
U2 - shared alpha	59.7	31	3.1	0.1	0	6.1
U2 - dose-specific alpha	58.8	31.5	3.1	0.1	0	6.5
U3 - shared alpha	58.4	32.3	2.7	0.1	0	6.5
U3 - dose-specific alpha	59.6	31.4	3	0.1	0	5.9
OBD selection (%)						
U2 - shared alpha	69.5	14.9	0.5	0.1	0	15
U2 - dose-specific alpha	69.5	14.4	0.8	0.1	0	15.2
U3 - shared alpha	71.1	13.9	0.2	0	0	14.8
U3 - dose-specific alpha	71.9	12.6	0.4	0	0	15.1
Mean administrations						
U2 - shared alpha	18.9	6.8	1.6	0.5	0	-
U2 - dose-specific alpha	18.9	6.8	1.7	0.5	0	-
U3 - shared alpha	19.2	6.6	1.5	0.5	0	-
U3 - dose-specific alpha	19.2	6.6	1.5	0.4	0	-
Mean IPDE administrations						
U2 - shared alpha	5.6	3.3	0.5	0.2	0	-
U2 - dose-specific alpha	5.6	3.2	0.6	0.2	0	-
U3 - shared alpha	5.7	3.2	0.5	0.2	0	-
U3 - dose-specific alpha	5.7	3.2	0.5	0.2	0	-
Mean trial duration (days)						Days
U2 - shared alpha						1002
U2 - dose-specific alpha						1000.7
U3 - shared alpha						1001.9
U3 - dose-specific alpha						999.4

Scenario 12						
DLT rate	0.06	0.09	0.18	0.29	0.44	
Efficacy rate	0.05	0.10	0.12	0.40	0.60	
Utility 2	-0	0	-0.1	0.1	0.2	
Utility 3	40.6	42.4	40	52.4	58.4	
MTD selection (%)						
U2 - shared alpha	1.2	5.4	32.9	50	10.5	0
U2 - dose-specific alpha	1.4	4.8	33.3	49.5	11	0
U3 - shared alpha	1.3	5.6	32.3	51.2	9.6	0
U3 - dose-specific alpha	1.3	5.7	32.1	51.3	9.6	0
OBD selection (%)						
U2 - shared alpha	12.8	16.2	16.2	43.3	9	2.5
U2 - dose-specific alpha	11.9	16.8	16.3	42.7	9.7	2.6
U3 - shared alpha	16.2	17.5	14.3	41.8	7.8	2.4
U3 - dose-specific alpha	16.6	17.7	13.4	41.8	8	2.5
Mean administrations						
U2 - shared alpha	5	6.3	6.5	9.1	2.9	-
U2 - dose-specific alpha	5	6.3	6.5	9	2.9	-
U3 - shared alpha	5.4	6.4	6.3	8.9	2.7	-
U3 - dose-specific alpha	5.4	6.4	6.3	8.9	2.7	-
Mean IPDE administrations						
U2 - shared alpha	0.8	3.8	2	3.3	0.9	-
U2 - dose-specific alpha	0.8	3.8	2	3.2	0.9	-
U3 - shared alpha	1	3.9	1.9	3.2	0.8	-
U3 - dose-specific alpha	1	3.9	1.9	3.2	0.8	-
Mean trial duration (days)						Days
U2 - shared alpha						1046.7
U2 - dose-specific alpha						1046.4
U3 - shared alpha						1041.5
U3 - dose-specific alpha						1041

Priors: efficacy beta-binomial Beta(0.5, 0.5); efficacy additive alpha Beta(0.15, 0.85); toxicity additive alpha Beta(0.15, 0.85).

Final column: early stopping % for MTD selection and no-OBD % for OBD selection; shaded rows are the dose-specific alpha efficacy model.

Phase I/II Operating Characteristics

Shared vs dose-specific additive-alpha efficacy - IPDE alpha = 0.9 - allocation = one-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No MTD/OBD %
Scenario 13						
DLT rate	0.21	0.29	0.30	0.39	0.46	
Efficacy rate	0.05	0.30	0.32	0.35	0.40	
Utility 2	-0.2	0	0	-0	-0.1	
Utility 3	34.6	46.4	47.2	45.4	45.6	
MTD selection (%)						
U2 - shared alpha	58.2	23	11.1	3.3	0.2	4.2
U2 - dose-specific alpha	59.2	21.7	11.5	3.1	0.3	4.2
U3 - shared alpha	58.7	23.6	10.8	2.8	0.1	4
U3 - dose-specific alpha	58.6	23.2	10.9	2.9	0.1	4.3
OBD selection (%)						
U2 - shared alpha	14.2	24.7	7.9	1.7	0.1	51.4
U2 - dose-specific alpha	14.9	23.5	8.2	1.5	0.2	51.7
U3 - shared alpha	14.9	25.7	6.4	1.3	0.1	51.6
U3 - dose-specific alpha	15.1	25.5	5.8	1.5	0.1	52
Mean administrations						
U2 - shared alpha	10.6	8.1	3	1.2	0.2	-
U2 - dose-specific alpha	10.7	8.1	3	1.2	0.2	-
U3 - shared alpha	10.7	8.3	2.9	1.1	0.1	-
U3 - dose-specific alpha	10.7	8.3	2.9	1.1	0.1	-
Mean IPDE administrations						
U2 - shared alpha	2.7	3.9	1	0.5	0.1	-
U2 - dose-specific alpha	2.7	3.9	1	0.5	0.1	-
U3 - shared alpha	2.8	4	0.9	0.5	0.1	-
U3 - dose-specific alpha	2.8	4	0.9	0.5	0.1	-
Mean trial duration (days)						Days
U2 - shared alpha						817.6
U2 - dose-specific alpha						818.3
U3 - shared alpha						818
U3 - dose-specific alpha						818.1

Scenario 14						
DLT rate	0.09	0.15	0.23	0.28	0.39	
Efficacy rate	0.05	0.08	0.12	0.30	0.40	
Utility 2	-0	-0.1	-0.1	0	0	
Utility 3	39.4	38.8	38	46.8	48.4	
MTD selection (%)						
U2 - shared alpha	5.6	15.3	37.5	31	10.6	0
U2 - dose-specific alpha	5.5	16.1	36.8	31.2	10.4	0
U3 - shared alpha	5.4	15.4	37.9	31.5	9.8	0
U3 - dose-specific alpha	5.7	15.9	37.1	31.2	10.1	0
OBD selection (%)						
U2 - shared alpha	20	21.3	20.3	23.7	5.9	8.8
U2 - dose-specific alpha	20.2	21.2	20.5	23.6	5.6	8.9
U3 - shared alpha	25	24	16	21.4	4.7	8.9
U3 - dose-specific alpha	26.1	22.8	16.5	21.1	4.5	9
Mean administrations						
U2 - shared alpha	6.7	7.3	6.6	6.4	2.2	-
U2 - dose-specific alpha	6.7	7.2	6.6	6.3	2.2	-
U3 - shared alpha	7.2	7.3	6.4	6.2	2.1	-
U3 - dose-specific alpha	7.1	7.3	6.4	6.2	2.1	-
Mean IPDE administrations						
U2 - shared alpha	1.5	4.1	2.2	2.5	0.7	-
U2 - dose-specific alpha	1.5	4.1	2.2	2.5	0.7	-
U3 - shared alpha	1.8	4.2	2.1	2.4	0.7	-
U3 - dose-specific alpha	1.7	4.2	2.1	2.5	0.7	-
Mean trial duration (days)						Days
U2 - shared alpha						988.1
U2 - dose-specific alpha						987.2
U3 - shared alpha						986.6
U3 - dose-specific alpha						984.6

Priors: efficacy beta-binomial Beta(0.5, 0.5); efficacy additive alpha Beta(0.15, 0.85); toxicity additive alpha Beta(0.15, 0.85).

Final column: early stopping % for MTD selection and no-OBD % for OBD selection; shaded rows are the dose-specific alpha efficacy model.

Phase I/II Operating Characteristics

Shared vs dose-specific additive-alpha efficacy - IPDE alpha = 0.9 - allocation = one-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No MTD/OBD %
Scenario 15						
DLT rate	0.13	0.20	0.32	0.39	0.46	
Efficacy rate	0.10	0.10	0.10	0.10	0.10	
Utility 2	-0	-0.1	-0.2	-0.3	-0.4	
Utility 3	40.8	38	33.2	30.4	27.6	
MTD selection (%)						
U2 - shared alpha	16.2	40.4	36	6.7	0.4	0.3
U2 - dose-specific alpha	15.8	40.9	35.9	6.6	0.5	0.3
U3 - shared alpha	15.9	41.7	35.4	6.3	0.4	0.3
U3 - dose-specific alpha	15.9	40.9	36.5	5.9	0.5	0.3
OBD selection (%)						
U2 - shared alpha	41.9	31.5	9.6	1.9	0.1	15
U2 - dose-specific alpha	43	30.5	10	1.8	0.1	14.6
U3 - shared alpha	45.5	30.3	7.4	1.2	0.1	15.5
U3 - dose-specific alpha	45.3	30.3	7.6	1.1	0.1	15.6
Mean administrations						
U2 - shared alpha	11.3	9	4.8	2.6	0.6	-
U2 - dose-specific alpha	11.2	9	4.9	2.7	0.7	-
U3 - shared alpha	11.8	8.8	4.6	2.6	0.6	-
U3 - dose-specific alpha	11.8	8.8	4.6	2.6	0.6	-
Mean IPDE administrations						
U2 - shared alpha	3.3	4.6	1.7	1.2	0.3	-
U2 - dose-specific alpha	3.3	4.6	1.7	1.2	0.3	-
U3 - shared alpha	3.6	4.5	1.6	1.1	0.3	-
U3 - dose-specific alpha	3.6	4.5	1.6	1.1	0.3	-
Mean trial duration (days)						Days
U2 - shared alpha						953
U2 - dose-specific alpha						955.3
U3 - shared alpha						952.2
U3 - dose-specific alpha						950.8

Scenario 16						
DLT rate	0.02	0.06	0.09	0.14	0.26	
Efficacy rate	0.20	0.20	0.20	0.20	0.20	
Utility 2	0.2	0.1	0.1	0.1	-0.1	
Utility 3	51.2	49.6	48.4	46.4	41.6	
MTD selection (%)						
U2 - shared alpha	0	1.4	4.9	18.7	75	0
U2 - dose-specific alpha	0	1.9	4.9	19.2	74	0
U3 - shared alpha	0	3.1	6.9	21.1	68.9	0
U3 - dose-specific alpha	0	3	7	21.5	68.5	0
OBD selection (%)						
U2 - shared alpha	31	30.6	16.4	13.8	8.2	0
U2 - dose-specific alpha	32	30.2	15.9	14.8	7.1	0
U3 - shared alpha	36.5	33.7	14.7	10.9	4.2	0
U3 - dose-specific alpha	37.3	33.1	13.4	11.4	4.8	0
Mean administrations						
U2 - shared alpha	4.4	5.6	6.2	8.1	5.7	-
U2 - dose-specific alpha	4.5	5.7	6.2	8.1	5.6	-
U3 - shared alpha	4.9	6.2	6.3	7.7	5	-
U3 - dose-specific alpha	4.8	6.2	6.4	7.7	4.9	-
Mean IPDE administrations						
U2 - shared alpha	0.6	3.1	2.4	3.5	2.2	-
U2 - dose-specific alpha	0.6	3.1	2.4	3.4	2.2	-
U3 - shared alpha	0.8	3.3	2.4	3.3	1.9	-
U3 - dose-specific alpha	0.8	3.3	2.5	3.3	1.9	-
Mean trial duration (days)						Days
U2 - shared alpha						1006.7
U2 - dose-specific alpha						1007.1
U3 - shared alpha						1004.7
U3 - dose-specific alpha						1004.4

Priors: efficacy beta-binomial Beta(0.5, 0.5); efficacy additive alpha Beta(0.15, 0.85); toxicity additive alpha Beta(0.15, 0.85).

Final column: early stopping % for MTD selection and no-OBD % for OBD selection; shaded rows are the dose-specific alpha efficacy model.

Phase I/II Operating Characteristics

Shared vs dose-specific additive-alpha efficacy - IPDE alpha = 0.9 - allocation = one-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No MTD/OBD %
Scenario 17						
DLT rate	0.11	0.23	0.39	0.58	0.76	
Efficacy rate	0.30	0.50	0.50	0.50	0.50	
Utility 2	0.2	0.3	0.1	-0.1	-0.3	
Utility 3	53.6	60.8	54.4	46.8	39.6	
MTD selection (%)						
U2 - shared alpha	10.8	78	10.9	0.2	0	0.1
U2 - dose-specific alpha	10.8	77.9	11.1	0.1	0	0.1
U3 - shared alpha	11.2	80.4	8.2	0.1	0	0.1
U3 - dose-specific alpha	11.3	80.9	7.6	0.1	0	0.1
OBD selection (%)						
U2 - shared alpha	22.5	70.1	7	0.1	0	0.3
U2 - dose-specific alpha	22.8	69.3	7.2	0.1	0	0.6
U3 - shared alpha	24.2	70.3	5.1	0.1	0	0.3
U3 - dose-specific alpha	25.4	69.3	4.9	0.1	0	0.3
Mean administrations						
U2 - shared alpha	10.9	16.6	2.3	0.2	0	-
U2 - dose-specific alpha	10.9	16.6	2.3	0.2	0	-
U3 - shared alpha	11.4	16.7	1.7	0.1	0	-
U3 - dose-specific alpha	11.5	16.6	1.7	0.1	0	-
Mean IPDE administrations						
U2 - shared alpha	2.5	5.6	0.7	0.1	0	-
U2 - dose-specific alpha	2.5	5.6	0.7	0.1	0	-
U3 - shared alpha	2.7	5.6	0.5	0.1	0	-
U3 - dose-specific alpha	2.8	5.6	0.5	0	0	-
Mean trial duration (days)						
U2 - shared alpha						Days
U2 - dose-specific alpha						1165.1
U3 - shared alpha						1162.8
U3 - dose-specific alpha						1162.1
						1161.1

Scenario 18						
DLT rate	0.02	0.04	0.10	0.20	0.35	
Efficacy rate	0.05	0.10	0.15	0.20	0.50	
Utility 2	0	0.1	0.1	0	0.1	
Utility 3	42.2	44.4	45	44	56	
MTD selection (%)						
U2 - shared alpha	0.1	0.2	4.5	37.9	57.3	0
U2 - dose-specific alpha	0.1	0.2	4.2	38.5	57	0
U3 - shared alpha	0.1	0.2	4.4	38	57.3	0
U3 - dose-specific alpha	0.1	0.2	4.2	38.6	56.9	0
OBD selection (%)						
U2 - shared alpha	9.3	12.1	13.4	20.1	44.9	0.2
U2 - dose-specific alpha	9.9	11.1	13.1	21.2	44.5	0.2
U3 - shared alpha	14.2	13.6	11.8	17.8	42.4	0.2
U3 - dose-specific alpha	13.6	15.1	13	17	41.2	0.1
Mean administrations						
U2 - shared alpha	3.5	4.2	5.4	7.9	8.9	-
U2 - dose-specific alpha	3.5	4.2	5.5	7.8	8.9	-
U3 - shared alpha	3.5	4.4	5.6	7.9	8.5	-
U3 - dose-specific alpha	3.6	4.4	5.6	7.9	8.5	-
Mean IPDE administrations						
U2 - shared alpha	0.2	2.9	1.9	3.2	2.7	-
U2 - dose-specific alpha	0.2	2.9	1.9	3.1	2.7	-
U3 - shared alpha	0.2	3	1.9	3.2	2.6	-
U3 - dose-specific alpha	0.3	3	1.9	3.2	2.6	-
Mean trial duration (days)						
U2 - shared alpha						Days
U2 - dose-specific alpha						1052
U3 - shared alpha						1050.6
U3 - dose-specific alpha						1046.9
						1046.7

Priors: efficacy beta-binomial Beta(0.5, 0.5); efficacy additive alpha Beta(0.15, 0.85); toxicity additive alpha Beta(0.15, 0.85).

Final column: early stopping % for MTD selection and no-OBD % for OBD selection; shaded rows are the dose-specific alpha efficacy model.

Phase I/II Operating Characteristics

Shared vs dose-specific additive-alpha efficacy - IPDE alpha = 0.9 - allocation = one-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No MTD/OBD %
Scenario 19						
DLT rate	0.11	0.26	0.39	0.60	0.77	
Efficacy rate	0.08	0.30	0.32	0.35	0.40	
Utility 2	-0	0	-0.1	-0.2	-0.4	
Utility 3	40.4	47.6	43.6	37	33.2	
MTD selection (%)						
U2 - shared alpha	26	56	17.4	0.6	0	0
U2 - dose-specific alpha	25.2	57.2	17	0.6	0	0
U3 - shared alpha	25	57.1	17.4	0.5	0	0
U3 - dose-specific alpha	26.5	56.1	16.7	0.7	0	0
OBD selection (%)						
U2 - shared alpha	17.2	55.1	9.8	0.2	0	17.7
U2 - dose-specific alpha	16.6	56.4	9.6	0.2	0	17.2
U3 - shared alpha	18.6	55.2	8.6	0.2	0	17.4
U3 - dose-specific alpha	19	54.5	7.9	0.3	0	18.3
Mean administrations						
U2 - shared alpha	9.6	13	4.6	1.1	0.1	-
U2 - dose-specific alpha	9.5	13.1	4.6	1	0.1	-
U3 - shared alpha	9.7	13.3	4.3	0.9	0.1	-
U3 - dose-specific alpha	9.7	13.3	4.2	1	0	-
Mean IPDE administrations						
U2 - shared alpha	2.3	5.7	1.5	0.5	0	-
U2 - dose-specific alpha	2.3	5.7	1.4	0.5	0	-
U3 - shared alpha	2.4	5.9	1.3	0.4	0	-
U3 - dose-specific alpha	2.4	5.9	1.3	0.4	0	-
Mean trial duration (days)						Days
U2 - shared alpha						1003.2
U2 - dose-specific alpha						1004.2
U3 - shared alpha						1000.6
U3 - dose-specific alpha						999.9

Scenario 20						
DLT rate	0.07	0.14	0.31	0.41	0.52	
Efficacy rate	0.05	0.25	0.25	0.25	0.25	
Utility 2	-0	0.1	-0.1	-0.2	-0.3	
Utility 3	40.2	49.4	42.6	38.6	34.2	
MTD selection (%)						
U2 - shared alpha	3.3	26.9	53.2	16	0.6	0
U2 - dose-specific alpha	3.1	27	54.3	15	0.6	0
U3 - shared alpha	2.8	27.1	53.8	15.7	0.6	0
U3 - dose-specific alpha	3	26.8	54.9	14.4	0.9	0
OBD selection (%)						
U2 - shared alpha	9	57	25.6	5.6	0	2.8
U2 - dose-specific alpha	8.2	56.3	27.1	5.4	0	3
U3 - shared alpha	11	57.9	22.8	5.6	0	2.7
U3 - dose-specific alpha	11.7	58.6	22.2	4.7	0.1	2.7
Mean administrations						
U2 - shared alpha	5.3	11.8	8.2	3.6	0.7	-
U2 - dose-specific alpha	5.3	11.9	8.1	3.6	0.7	-
U3 - shared alpha	5.6	12.4	7.7	3.3	0.6	-
U3 - dose-specific alpha	5.6	12.4	7.7	3.2	0.6	-
Mean IPDE administrations						
U2 - shared alpha	0.9	5.5	2.7	1.4	0.3	-
U2 - dose-specific alpha	0.9	5.5	2.6	1.4	0.3	-
U3 - shared alpha	1	5.7	2.5	1.3	0.2	-
U3 - dose-specific alpha	1	5.8	2.5	1.3	0.2	-
Mean trial duration (days)						Days
U2 - shared alpha						1037.3
U2 - dose-specific alpha						1035.9
U3 - shared alpha						1032.9
U3 - dose-specific alpha						1032.6

Priors: efficacy beta-binomial Beta(0.5, 0.5); efficacy additive alpha Beta(0.15, 0.85); toxicity additive alpha Beta(0.15, 0.85).

Final column: early stopping % for MTD selection and no-OBD % for OBD selection; shaded rows are the dose-specific alpha efficacy model.

Phase I/II Operating Characteristics

Shared vs dose-specific additive-alpha efficacy - IPDE alpha = 0.9 - allocation = one-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No MTD/OBD %
Scenario 21						
DLT rate	0.37	0.56	0.69	0.83	0.88	
Efficacy rate	0.50	0.50	0.50	0.50	0.50	
Utility 2	0.1	-0.1	-0.2	-0.3	-0.4	
Utility 3	55.2	47.6	42.4	36.8	34.8	
MTD selection (%)						
U2 - shared alpha	40.4	0.4	0	0	0	59.2
U2 - dose-specific alpha	40.4	0.4	0	0	0	59.2
U3 - shared alpha	41.1	0.4	0	0	0	58.5
U3 - dose-specific alpha	40.4	0.4	0	0	0	59.2
OBD selection (%)						
U2 - shared alpha	37	0.1	0	0	0	62.9
U2 - dose-specific alpha	37.1	0.1	0	0	0	62.8
U3 - shared alpha	37.7	0	0	0	0	62.3
U3 - dose-specific alpha	37	0.1	0	0	0	62.9
Mean administrations						
U2 - shared alpha	18.2	1.7	0	0	0	-
U2 - dose-specific alpha	18.2	1.7	0	0	0	-
U3 - shared alpha	18.3	1.6	0	0	0	-
U3 - dose-specific alpha	18.3	1.6	0	0	0	-
Mean IPDE administrations						
U2 - shared alpha	3.5	0.8	0	0	0	-
U2 - dose-specific alpha	3.5	0.8	0	0	0	-
U3 - shared alpha	3.5	0.8	0	0	0	-
U3 - dose-specific alpha	3.5	0.8	0	0	0	-
Mean trial duration (days)						Days
U2 - shared alpha						858.9
U2 - dose-specific alpha						858.7
U3 - shared alpha						861.1
U3 - dose-specific alpha						859.9

Scenario 22						
DLT rate	0.07	0.27	0.50	0.56	0.78	
Efficacy rate	0.08	0.25	0.25	0.50	0.50	
Utility 2	0	-0	-0.2	-0.1	-0.3	
Utility 3	42	44.2	35	47.6	38.8	
MTD selection (%)						
U2 - shared alpha	21.7	62.6	15.5	0.2	0	0
U2 - dose-specific alpha	20.8	63.3	15.7	0.2	0	0
U3 - shared alpha	21.1	61.5	17.1	0.3	0	0
U3 - dose-specific alpha	19.8	62.1	17.7	0.4	0	0
OBD selection (%)						
U2 - shared alpha	19.8	58.9	6.9	0.2	0	14.2
U2 - dose-specific alpha	19.8	59.5	6.9	0.2	0	13.6
U3 - shared alpha	21.5	57.4	6.8	0.3	0	14
U3 - dose-specific alpha	19.8	59.6	6.6	0.4	0	13.6
Mean administrations						
U2 - shared alpha	9.7	13.5	4.3	1.1	0	-
U2 - dose-specific alpha	9.8	13.6	4.3	1.1	0	-
U3 - shared alpha	10	13.5	4.1	1	0	-
U3 - dose-specific alpha	10	13.5	4.1	1.1	0	-
Mean IPDE administrations						
U2 - shared alpha	2.5	6.1	1.3	0.5	0	-
U2 - dose-specific alpha	2.5	6.1	1.3	0.5	0	-
U3 - shared alpha	2.6	6.1	1.2	0.4	0	-
U3 - dose-specific alpha	2.6	6.1	1.2	0.4	0	-
Mean trial duration (days)						Days
U2 - shared alpha						1006.4
U2 - dose-specific alpha						1007.9
U3 - shared alpha						1002.1
U3 - dose-specific alpha						1001.6

Priors: efficacy beta-binomial Beta(0.5, 0.5); efficacy additive alpha Beta(0.15, 0.85); toxicity additive alpha Beta(0.15, 0.85).

Final column: early stopping % for MTD selection and no-OBD % for OBD selection; shaded rows are the dose-specific alpha efficacy model.

Phase I/II Operating Characteristics

Shared vs dose-specific additive-alpha efficacy - IPDE alpha = 0.9 - allocation = one-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No MTD/OBD %
Scenario 23						
DLT rate	0.32	0.55	0.80	0.94	0.97	
Efficacy rate	0.05	0.05	0.05	0.05	0.05	
Utility 2	-0.3	-0.5	-0.8	-0.9	-0.9	
Utility 3	30.2	21	11	5.4	4.2	
MTD selection (%)						
U2 - shared alpha	60.6	0.3	0	0	0	39.1
U2 - dose-specific alpha	60.7	0.3	0	0	0	39
U3 - shared alpha	61.5	0.4	0	0	0	38.1
U3 - dose-specific alpha	61.1	0.5	0	0	0	38.4
OBD selection (%)						
U2 - shared alpha	5.3	0.1	0	0	0	94.6
U2 - dose-specific alpha	5.5	0.1	0	0	0	94.4
U3 - shared alpha	5.6	0.1	0	0	0	94.3
U3 - dose-specific alpha	5.5	0.2	0	0	0	94.3
Mean administrations						
U2 - shared alpha	10.7	3.2	0.2	0	0	-
U2 - dose-specific alpha	10.8	3.2	0.2	0	0	-
U3 - shared alpha	10.8	3.2	0.2	0	0	-
U3 - dose-specific alpha	10.8	3.2	0.2	0	0	-
Mean IPDE administrations						
U2 - shared alpha	2.8	1.9	0.1	0	0	-
U2 - dose-specific alpha	2.9	1.9	0.1	0	0	-
U3 - shared alpha	2.9	1.9	0.1	0	0	-
U3 - dose-specific alpha	2.9	1.9	0.1	0	0	-
Mean trial duration (days)						Days
U2 - shared alpha						501.7
U2 - dose-specific alpha						504.2
U3 - shared alpha						505
U3 - dose-specific alpha						504.8

Scenario 24						
DLT rate	0.01	0.02	0.05	0.10	0.25	
Efficacy rate	0.05	0.08	0.20	0.50	0.50	
Utility 2	0	0.1	0.1	0.4	0.2	
Utility 3	42.6	44	50	66	60	
MTD selection (%)						
U2 - shared alpha	0	0	0.2	54.8	45	0
U2 - dose-specific alpha	0	0.1	0.2	54.1	45.6	0
U3 - shared alpha	0	0	0.3	58.9	40.8	0
U3 - dose-specific alpha	0	0.1	0.3	59.2	40.4	0
OBD selection (%)						
U2 - shared alpha	0.2	0.5	1.6	64.6	33.1	0
U2 - dose-specific alpha	0.4	0.5	1.4	63.8	33.9	0
U3 - shared alpha	0.7	1	1.8	68.3	28.2	0
U3 - dose-specific alpha	1.1	1.1	2.3	67.4	28.1	0
Mean administrations						
U2 - shared alpha	3.1	3.2	3.8	14	5.9	-
U2 - dose-specific alpha	3.1	3.2	3.8	13.9	6	-
U3 - shared alpha	3.1	3.2	3.9	14.5	5.3	-
U3 - dose-specific alpha	3.1	3.2	3.9	14.5	5.3	-
Mean IPDE administrations						
U2 - shared alpha	0	2.4	1.2	4.8	1.8	-
U2 - dose-specific alpha	0	2.4	1.2	4.8	1.8	-
U3 - shared alpha	0	2.5	1.2	5	1.6	-
U3 - dose-specific alpha	0	2.5	1.3	5	1.6	-
Mean trial duration (days)						Days
U2 - shared alpha						1087.8
U2 - dose-specific alpha						1087.6
U3 - shared alpha						1085.7
U3 - dose-specific alpha						1085.3

Priors: efficacy beta-binomial Beta(0.5, 0.5); efficacy additive alpha Beta(0.15, 0.85); toxicity additive alpha Beta(0.15, 0.85).

Final column: early stopping % for MTD selection and no-OBD % for OBD selection; shaded rows are the dose-specific alpha efficacy model.

Phase I/II Operating Characteristics

Shared vs dose-specific additive-alpha efficacy - IPDE alpha = 0.9 - allocation = one-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No MTD/OBD %
Scenario 25						
DLT rate	0.09	0.18	0.37	0.63	0.77	
Efficacy rate	0.20	0.25	0.40	0.45	0.50	
Utility 2	0.1	0.1	0	-0.2	-0.3	
Utility 3	48.4	47.8	49.2	41.8	39.2	
MTD selection (%)						
U2 - shared alpha	5.2	49.2	43.3	2.3	0	0
U2 - dose-specific alpha	5	49.1	44.3	1.6	0	0
U3 - shared alpha	4.8	48.8	44.8	1.6	0	0
U3 - dose-specific alpha	4.7	50.9	42.4	2	0	0
OBD selection (%)						
U2 - shared alpha	28.8	40.6	28.4	1.2	0	1
U2 - dose-specific alpha	27.9	41.4	28.5	1	0	1.2
U3 - shared alpha	32.9	40.3	25.2	0.6	0	1
U3 - dose-specific alpha	33.1	42	22.8	0.9	0	1.2
Mean administrations						
U2 - shared alpha	9.5	11.2	7.7	1.4	0.1	-
U2 - dose-specific alpha	9.6	11.3	7.6	1.3	0.1	-
U3 - shared alpha	10.2	11.4	7.2	1.1	0	-
U3 - dose-specific alpha	10.3	11.3	7	1.2	0	-
Mean IPDE administrations						
U2 - shared alpha	2.4	4.7	2.4	0.5	0	-
U2 - dose-specific alpha	2.4	4.8	2.4	0.5	0	-
U3 - shared alpha	2.7	4.9	2.2	0.5	0	-
U3 - dose-specific alpha	2.8	4.8	2.2	0.5	0	-
Mean trial duration (days)						Days
U2 - shared alpha						1090.1
U2 - dose-specific alpha						1088.3
U3 - shared alpha						1084
U3 - dose-specific alpha						1082.1

Scenario 26						
DLT rate	0.15	0.22	0.30	0.62	0.80	
Efficacy rate	0.10	0.20	0.35	0.45	0.55	
Utility 2	-0.1	-0	0.1	-0.2	-0.2	
Utility 3	40	43.2	49	42.2	41	
MTD selection (%)						
U2 - shared alpha	25.2	43.7	28	2.8	0	0.3
U2 - dose-specific alpha	25.4	43	28.5	2.8	0	0.3
U3 - shared alpha	24.9	44.4	27.2	3.2	0	0.3
U3 - dose-specific alpha	25.2	43.8	27.9	2.8	0	0.3
OBD selection (%)						
U2 - shared alpha	26.5	35.4	20.9	1.6	0	15.6
U2 - dose-specific alpha	25.6	36.5	20.5	1.8	0	15.6
U3 - shared alpha	29	34.8	19.2	1.6	0	15.4
U3 - dose-specific alpha	28.1	36.4	18.4	1.5	0	15.6
Mean administrations						
U2 - shared alpha	10.4	9.8	6.2	1.6	0.1	-
U2 - dose-specific alpha	10.4	9.9	6.2	1.6	0.1	-
U3 - shared alpha	10.7	9.8	6	1.5	0.1	-
U3 - dose-specific alpha	10.7	10	5.9	1.5	0	-
Mean IPDE administrations						
U2 - shared alpha	2.8	4.7	1.9	0.7	0	-
U2 - dose-specific alpha	2.8	4.7	1.9	0.7	0	-
U3 - shared alpha	2.9	4.7	1.8	0.6	0	-
U3 - dose-specific alpha	2.9	4.7	1.8	0.7	0	-
Mean trial duration (days)						Days
U2 - shared alpha						991.3
U2 - dose-specific alpha						990.3
U3 - shared alpha						989.8
U3 - dose-specific alpha						989.9

Priors: efficacy beta-binomial Beta(0.5, 0.5); efficacy additive alpha Beta(0.15, 0.85); toxicity additive alpha Beta(0.15, 0.85).

Final column: early stopping % for MTD selection and no-OBD % for OBD selection; shaded rows are the dose-specific alpha efficacy model.

Phase I/II Operating Characteristics

Shared vs dose-specific additive-alpha efficacy - IPDE alpha = 0.9 - allocation = one-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No MTD/OBD %
Scenario 27						
DLT rate	0.11	0.17	0.30	0.50	0.62	
Efficacy rate	0.10	0.20	0.40	0.42	0.45	
Utility 2	-0	0	0.1	-0.1	-0.2	
Utility 3	41.6	45.2	52	45.2	42.2	
MTD selection (%)						
U2 - shared alpha	8	36.4	49	6.6	0	0
U2 - dose-specific alpha	8.8	36.4	47.7	7.1	0	0
U3 - shared alpha	8.3	36.9	48.3	6.5	0	0
U3 - dose-specific alpha	8.4	37	47.9	6.7	0	0
OBD selection (%)						
U2 - shared alpha	17.2	33.7	39	3.7	0	6.4
U2 - dose-specific alpha	16.5	34.1	39.1	3.7	0	6.6
U3 - shared alpha	19.3	34.4	36.7	2.9	0	6.7
U3 - dose-specific alpha	19.9	34.4	35.8	3.1	0	6.8
Mean administrations						
U2 - shared alpha	7.7	9.7	9.4	2.3	0.2	-
U2 - dose-specific alpha	7.6	9.8	9.3	2.3	0.2	-
U3 - shared alpha	8	9.9	9.1	2.2	0.2	-
U3 - dose-specific alpha	8	10	9	2.1	0.1	-
Mean IPDE administrations						
U2 - shared alpha	1.8	4.5	2.8	0.9	0.1	-
U2 - dose-specific alpha	1.8	4.6	2.8	0.9	0.1	-
U3 - shared alpha	1.9	4.6	2.7	0.9	0.1	-
U3 - dose-specific alpha	2	4.7	2.7	0.8	0.1	-
Mean trial duration (days)						Days
U2 - shared alpha						1051.4
U2 - dose-specific alpha						1051.2
U3 - shared alpha						1046.3
U3 - dose-specific alpha						1045.2

Scenario 28						
DLT rate	0.20	0.30	0.50	0.60	0.77	
Efficacy rate	0.05	0.30	0.32	0.35	0.40	
Utility 2	-0.1	0	-0.2	-0.2	-0.4	
Utility 3	35	46	39.2	37	33.2	
MTD selection (%)						
U2 - shared alpha	68.6	24.6	2.8	0.1	0	3.9
U2 - dose-specific alpha	67.7	26	2.5	0.1	0	3.7
U3 - shared alpha	67.3	25.4	3.5	0.1	0	3.7
U3 - dose-specific alpha	67	25.8	3.1	0.2	0	3.9
OBD selection (%)						
U2 - shared alpha	18.4	23.4	1.5	0	0	56.7
U2 - dose-specific alpha	18.2	24.2	1.3	0	0	56.3
U3 - shared alpha	17.7	24.3	1.5	0	0	56.5
U3 - dose-specific alpha	18.2	24.4	1.4	0	0	56
Mean administrations						
U2 - shared alpha	11.7	8.8	2.1	0.4	0	-
U2 - dose-specific alpha	11.6	8.9	2.1	0.4	0	-
U3 - shared alpha	11.6	8.9	2	0.4	0	-
U3 - dose-specific alpha	11.7	8.9	2	0.4	0	-
Mean IPDE administrations						
U2 - shared alpha	3.1	4.1	0.7	0.2	0	-
U2 - dose-specific alpha	3.1	4.2	0.7	0.2	0	-
U3 - shared alpha	3.1	4.2	0.6	0.2	0	-
U3 - dose-specific alpha	3.1	4.2	0.6	0.2	0	-
Mean trial duration (days)						Days
U2 - shared alpha						814.2
U2 - dose-specific alpha						814.9
U3 - shared alpha						813.3
U3 - dose-specific alpha						812.4

Priors: efficacy beta-binomial Beta(0.5, 0.5); efficacy additive alpha Beta(0.15, 0.85); toxicity additive alpha Beta(0.15, 0.85).

Final column: early stopping % for MTD selection and no-OBD % for OBD selection; shaded rows are the dose-specific alpha efficacy model.

Phase I/II Operating Characteristics

Shared vs dose-specific additive-alpha efficacy - IPDE alpha = 0.9 - allocation = one-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No MTD/OBD %
Scenario 29						
DLT rate	0.19	0.30	0.56	0.74	0.86	
Efficacy rate	0.40	0.40	0.40	0.40	0.40	
Utility 2	0.2	0.1	-0.2	-0.3	-0.5	
Utility 3	56.4	52	41.6	34.4	29.6	
MTD selection (%)						
U2 - shared alpha	50.6	43.6	2.4	0	0	3.4
U2 - dose-specific alpha	50.4	44	2.6	0	0	3
U3 - shared alpha	51.1	43.8	1.7	0	0	3.4
U3 - dose-specific alpha	50.6	44.6	1.6	0	0	3.2
OBD selection (%)						
U2 - shared alpha	69.9	25.6	0.7	0	0	3.8
U2 - dose-specific alpha	70.5	25.1	1	0	0	3.4
U3 - shared alpha	72.9	23.2	0.5	0	0	3.4
U3 - dose-specific alpha	72.2	23.5	0.7	0	0	3.6
Mean administrations						
U2 - shared alpha	20.1	8.2	0.9	0.1	0	-
U2 - dose-specific alpha	20.2	8.2	0.9	0.1	0	-
U3 - shared alpha	20.7	8	0.6	0.1	0	-
U3 - dose-specific alpha	20.7	8	0.6	0	0	-
Mean IPDE administrations						
U2 - shared alpha	5	3.1	0.3	0	0	-
U2 - dose-specific alpha	5	3.1	0.3	0	0	-
U3 - shared alpha	5.2	3.1	0.2	0	0	-
U3 - dose-specific alpha	5.3	3.1	0.2	0	0	-
Mean trial duration (days)						Days
U2 - shared alpha						1149.2
U2 - dose-specific alpha						1151
U3 - shared alpha						1149.4
U3 - dose-specific alpha						1148.1

Scenario 30						
DLT rate	0.05	0.08	0.17	0.27	0.54	
Efficacy rate	0.12	0.15	0.20	0.40	0.50	
Utility 2	0.1	0.1	0	0.1	-0	
Utility 3	45.2	45.8	45.2	53.2	48.4	
MTD selection (%)						
U2 - shared alpha	0.5	3.2	29	57.3	10	0
U2 - dose-specific alpha	0.5	3	28.9	58	9.6	0
U3 - shared alpha	0.5	3.4	28.3	59.4	8.4	0
U3 - dose-specific alpha	0.5	3.5	27.9	58.9	9.2	0
OBD selection (%)						
U2 - shared alpha	14.3	17.2	18.6	43.1	6.4	0.4
U2 - dose-specific alpha	14.2	16.1	19.1	43.9	6.2	0.5
U3 - shared alpha	19.5	18.9	16.1	40.6	4.4	0.5
U3 - dose-specific alpha	19.7	19.2	16.1	39.7	4.8	0.5
Mean administrations						
U2 - shared alpha	4.7	6.1	7	9.9	2.3	-
U2 - dose-specific alpha	4.7	6.1	6.9	9.9	2.4	-
U3 - shared alpha	4.8	6.5	6.9	9.7	2.1	-
U3 - dose-specific alpha	4.9	6.4	6.9	9.7	2.1	-
Mean IPDE administrations						
U2 - shared alpha	0.7	3.4	2.3	3.4	0.8	-
U2 - dose-specific alpha	0.7	3.4	2.3	3.4	0.8	-
U3 - shared alpha	0.8	3.6	2.3	3.3	0.7	-
U3 - dose-specific alpha	0.8	3.6	2.3	3.3	0.7	-
Mean trial duration (days)						Days
U2 - shared alpha						1064.2
U2 - dose-specific alpha						1065
U3 - shared alpha						1060.5
U3 - dose-specific alpha						1060

Priors: efficacy beta-binomial Beta(0.5, 0.5); efficacy additive alpha Beta(0.15, 0.85); toxicity additive alpha Beta(0.15, 0.85).

Final column: early stopping % for MTD selection and no-OBD % for OBD selection; shaded rows are the dose-specific alpha efficacy model.

Phase I/II Operating Characteristics

Shared vs dose-specific additive-alpha efficacy - IPDE alpha = 0.9 - allocation = one-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No MTD/OBD %
Scenario 31						
DLT rate	0.04	0.06	0.10	0.20	0.60	
Efficacy rate	0.20	0.50	0.50	0.50	0.50	
Utility 2	0.2	0.4	0.4	0.3	-0.1	
Utility 3	50.4	67.6	66	62	46	
MTD selection (%)						
U2 - shared alpha	0.1	52.5	31.4	14.4	1.6	0
U2 - dose-specific alpha	0.1	52	32.5	14	1.4	0
U3 - shared alpha	0.1	58.7	28.4	11.5	1.3	0
U3 - dose-specific alpha	0.1	57.8	29	12	1.1	0
OBD selection (%)						
U2 - shared alpha	1.5	54.5	31.8	11.6	0.6	0
U2 - dose-specific alpha	1.4	54.3	32.1	11.7	0.5	0
U3 - shared alpha	2.3	59.9	27.7	9.6	0.5	0
U3 - dose-specific alpha	2	59.9	27.9	9.8	0.4	0
Mean administrations						
U2 - shared alpha	4.1	16.2	7.3	2.3	0.2	-
U2 - dose-specific alpha	4.1	16.3	7.3	2.2	0.2	-
U3 - shared alpha	4.1	17.3	6.5	1.9	0.1	-
U3 - dose-specific alpha	4.2	17.1	6.6	2	0.1	-
Mean IPDE administrations						
U2 - shared alpha	0.4	6	2.3	0.7	0.1	-
U2 - dose-specific alpha	0.4	6.1	2.3	0.7	0.1	-
U3 - shared alpha	0.4	6.4	2	0.6	0	-
U3 - dose-specific alpha	0.4	6.3	2.1	0.6	0	-
Mean trial duration (days)						Days
U2 - shared alpha						1132.7
U2 - dose-specific alpha						1132.2
U3 - shared alpha						1130.7
U3 - dose-specific alpha						1130.7

Scenario 32						
DLT rate	0.02	0.05	0.11	0.30	0.36	
Efficacy rate	0.08	0.15	0.30	0.30	0.30	
Utility 2	0.1	0.1	0.2	0	-0.1	
Utility 3	44	47	53.6	46	43.6	
MTD selection (%)						
U2 - shared alpha	0.1	0.3	14.1	56.8	28.7	0
U2 - dose-specific alpha	0.1	0.3	13.6	59.2	26.8	0
U3 - shared alpha	0.1	0.5	15.4	59.5	24.5	0
U3 - dose-specific alpha	0.1	0.1	15.5	60.6	23.7	0
OBD selection (%)						
U2 - shared alpha	9.2	11.5	35.7	35.3	8.1	0.2
U2 - dose-specific alpha	9.5	13.7	35	33.7	7.9	0.2
U3 - shared alpha	13	17.3	36.4	28.5	4.7	0.1
U3 - dose-specific alpha	13.6	18	34.5	28.6	5.2	0.1
Mean administrations						
U2 - shared alpha	3.5	4.7	9.1	9	3.6	-
U2 - dose-specific alpha	3.5	4.6	9.1	9.1	3.6	-
U3 - shared alpha	3.6	4.9	9.5	8.9	3.1	-
U3 - dose-specific alpha	3.6	5	9.5	8.8	3	-
Mean IPDE administrations						
U2 - shared alpha	0.2	3	3	3.4	1.3	-
U2 - dose-specific alpha	0.2	3	3	3.4	1.3	-
U3 - shared alpha	0.3	3.1	3.2	3.3	1.1	-
U3 - dose-specific alpha	0.3	3.1	3.2	3.3	1	-
Mean trial duration (days)						Days
U2 - shared alpha						1049.7
U2 - dose-specific alpha						1050.3
U3 - shared alpha						1048
U3 - dose-specific alpha						1046.3

Priors: efficacy beta-binomial Beta(0.5, 0.5); efficacy additive alpha Beta(0.15, 0.85); toxicity additive alpha Beta(0.15, 0.85).

Final column: early stopping % for MTD selection and no-OBD % for OBD selection; shaded rows are the dose-specific alpha efficacy model.

Phase I/II Operating Characteristics

Shared vs dose-specific additive-alpha efficacy - IPDE alpha = 0.9 - allocation = one-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No MTD/OBD %
Scenario 33						
DLT rate	0.01	0.02	0.09	0.32	0.38	
Efficacy rate	0.05	0.08	0.20	0.40	0.50	
Utility 2	0	0.1	0.1	0.1	0.1	
Utility 3	42.6	44	48.4	51.2	54.8	
MTD selection (%)						
U2 - shared alpha	0	0	14.7	65.7	19.6	0
U2 - dose-specific alpha	0	0.1	14.4	66.1	19.4	0
U3 - shared alpha	0	0	13.9	68.5	17.6	0
U3 - dose-specific alpha	0	0	13.9	68.3	17.8	0
OBD selection (%)						
U2 - shared alpha	6.6	7.9	16.3	53.2	15.9	0.1
U2 - dose-specific alpha	8.8	6.7	15.9	53	15.5	0.1
U3 - shared alpha	12.6	7.9	13.9	52.7	12.8	0.1
U3 - dose-specific alpha	12.7	9	14.6	50.9	12.7	0.1
Mean administrations						
U2 - shared alpha	3.3	3.9	6.6	11.7	4.5	-
U2 - dose-specific alpha	3.3	3.9	6.6	11.6	4.5	-
U3 - shared alpha	3.4	4	6.7	11.8	4.1	-
U3 - dose-specific alpha	3.4	4	6.7	11.8	4.1	-
Mean IPDE administrations						
U2 - shared alpha	0.1	2.8	2	4.1	1.3	-
U2 - dose-specific alpha	0.2	2.8	1.9	4.1	1.3	-
U3 - shared alpha	0.2	2.8	2	4.2	1.2	-
U3 - dose-specific alpha	0.2	2.8	2	4.2	1.2	-
Mean trial duration (days)						Days
U2 - shared alpha						1081.5
U2 - dose-specific alpha						1080.9
U3 - shared alpha						1078.7
U3 - dose-specific alpha						1077.7

Scenario 34						
DLT rate	0.05	0.13	0.30	0.36	0.42	
Efficacy rate	0.05	0.10	0.30	0.32	0.35	
Utility 2	0	-0	0	-0	-0.1	
Utility 3	41	40.8	46	44.8	44.2	
MTD selection (%)						
U2 - shared alpha	2.2	17.8	56.3	21.1	2.6	0
U2 - dose-specific alpha	2.1	20.2	53.5	21.4	2.8	0
U3 - shared alpha	2.2	18.8	53.5	23.2	2.3	0
U3 - dose-specific alpha	2.5	17.6	54.4	23.3	2.2	0
OBD selection (%)						
U2 - shared alpha	13.2	18.8	49.4	12.5	1.5	4.6
U2 - dose-specific alpha	14.4	19.4	47.6	12.5	1.6	4.5
U3 - shared alpha	17.4	20.9	43.6	12.4	0.9	4.8
U3 - dose-specific alpha	17.7	20.1	43.6	12.7	1	4.9
Mean administrations						
U2 - shared alpha	5.6	7.7	10.2	5	1.1	-
U2 - dose-specific alpha	5.6	7.7	10.3	5	1.1	-
U3 - shared alpha	6	7.8	10.1	4.8	0.9	-
U3 - dose-specific alpha	5.9	7.8	10.1	4.8	0.9	-
Mean IPDE administrations						
U2 - shared alpha	1.1	4.1	3.3	1.9	0.4	-
U2 - dose-specific alpha	1.1	4.1	3.3	1.9	0.4	-
U3 - shared alpha	1.3	4.2	3.2	1.8	0.3	-
U3 - dose-specific alpha	1.3	4.1	3.2	1.8	0.4	-
Mean trial duration (days)						Days
U2 - shared alpha						1037
U2 - dose-specific alpha						1037.3
U3 - shared alpha						1033
U3 - dose-specific alpha						1034.6

Priors: efficacy beta-binomial Beta(0.5, 0.5); efficacy additive alpha Beta(0.15, 0.85); toxicity additive alpha Beta(0.15, 0.85).

Final column: early stopping % for MTD selection and no-OBD % for OBD selection; shaded rows are the dose-specific alpha efficacy model.

Phase I/II Operating Characteristics

Shared vs dose-specific additive-alpha efficacy - IPDE alpha = 0.9 - allocation = one-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No MTD/OBD %
Scenario 35						
DLT rate	0.03	0.09	0.30	0.40	0.45	
Efficacy rate	0.20	0.45	0.45	0.45	0.45	
Utility 2	0.2	0.4	0.1	0.1	0	
Utility 3	50.8	63.4	55	51	49	
MTD selection (%)						
U2 - shared alpha	0.1	42	48.3	9.3	0.3	0
U2 - dose-specific alpha	0.1	43.4	46.5	9.7	0.3	0
U3 - shared alpha	0.2	49.6	42.2	7.9	0.1	0
U3 - dose-specific alpha	0.2	48.5	43.1	8.2	0	0
OBD selection (%)						
U2 - shared alpha	4.9	55.1	35.3	4.4	0.2	0.1
U2 - dose-specific alpha	5	57.6	32.8	4.3	0.2	0.1
U3 - shared alpha	6.4	61.7	28.3	3.5	0	0.1
U3 - dose-specific alpha	6.5	62.9	26.8	3.7	0	0.1
Mean administrations						
U2 - shared alpha	4.4	15.5	8.2	1.7	0.2	-
U2 - dose-specific alpha	4.5	15.7	8	1.7	0.2	-
U3 - shared alpha	4.8	16.5	7.2	1.4	0.1	-
U3 - dose-specific alpha	4.8	16.5	7.2	1.4	0.1	-
Mean IPDE administrations						
U2 - shared alpha	0.5	5.8	2.4	0.6	0.1	-
U2 - dose-specific alpha	0.5	5.9	2.3	0.6	0.1	-
U3 - shared alpha	0.6	6.3	2.1	0.5	0	-
U3 - dose-specific alpha	0.7	6.3	2.1	0.5	0	-
Mean trial duration (days)						Days
U2 - shared alpha						1135.7
U2 - dose-specific alpha						1134.6
U3 - shared alpha						1129.6
U3 - dose-specific alpha						1129.4

Scenario 36						
DLT rate	0.13	0.29	0.35	0.40	0.43	
Efficacy rate	0.05	0.20	0.22	0.30	0.35	
Utility 2	-0.1	-0.1	-0.1	-0.1	-0.1	
Utility 3	37.8	40.4	39.2	42	43.8	
MTD selection (%)						
U2 - shared alpha	39.5	39.5	17.2	3.2	0.4	0.2
U2 - dose-specific alpha	40.1	38.9	17.4	3.2	0.2	0.2
U3 - shared alpha	39.8	39.7	16.9	3.2	0.2	0.2
U3 - dose-specific alpha	38.8	41.2	16.4	3.1	0.3	0.2
OBD selection (%)						
U2 - shared alpha	16.9	36.7	9.5	2.1	0.3	34.5
U2 - dose-specific alpha	17	36	10.1	2.1	0.1	34.7
U3 - shared alpha	20.1	35.9	8	1.7	0.1	34.2
U3 - dose-specific alpha	19.5	37.4	7.3	1.5	0.2	34.1
Mean administrations						
U2 - shared alpha	9.8	9.9	4.2	1.9	0.3	-
U2 - dose-specific alpha	9.6	10	4.3	1.9	0.3	-
U3 - shared alpha	10	10	4.1	1.8	0.3	-
U3 - dose-specific alpha	9.9	10.1	4.1	1.8	0.3	-
Mean IPDE administrations						
U2 - shared alpha	2.5	4.9	1.4	0.8	0.1	-
U2 - dose-specific alpha	2.4	5	1.4	0.8	0.1	-
U3 - shared alpha	2.5	5	1.3	0.7	0.1	-
U3 - dose-specific alpha	2.5	5	1.3	0.7	0.1	-
Mean trial duration (days)						Days
U2 - shared alpha						904.5
U2 - dose-specific alpha						903.5
U3 - shared alpha						900.3
U3 - dose-specific alpha						901.2

Priors: efficacy beta-binomial Beta(0.5, 0.5); efficacy additive alpha Beta(0.15, 0.85); toxicity additive alpha Beta(0.15, 0.85).

Final column: early stopping % for MTD selection and no-OBD % for OBD selection; shaded rows are the dose-specific alpha efficacy model.

Phase I/II Operating Characteristics

Shared vs dose-specific additive-alpha efficacy - IPDE alpha = 0.9 - allocation = one-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No MTD/OBD %
Scenario 37						
DLT rate	0.08	0.30	0.40	0.45	0.50	
Efficacy rate	0.05	0.20	0.20	0.20	0.20	
Utility 2	-0	-0.1	-0.2	-0.2	-0.3	
Utility 3	39.8	40	36	34	32	
MTD selection (%)						
U2 - shared alpha	31.1	51.3	16.6	1	0	0
U2 - dose-specific alpha	31.1	51.4	16.3	1.2	0	0
U3 - shared alpha	30.2	51.4	17.2	1.2	0	0
U3 - dose-specific alpha	31	50.1	17.7	1.2	0	0
OBD selection (%)						
U2 - shared alpha	18.1	47	7.8	0.2	0	26.9
U2 - dose-specific alpha	17.6	47.3	7.9	0.3	0	26.9
U3 - shared alpha	20.2	45.9	7.1	0.4	0	26.4
U3 - dose-specific alpha	20.6	44.5	7.1	0.4	0	27.4
Mean administrations						
U2 - shared alpha	9.4	11.6	4.4	1.7	0.3	-
U2 - dose-specific alpha	9.4	11.6	4.4	1.7	0.2	-
U3 - shared alpha	9.7	11.6	4.2	1.6	0.2	-
U3 - dose-specific alpha	9.7	11.6	4.2	1.7	0.2	-
Mean IPDE administrations						
U2 - shared alpha	2.3	5.7	1.4	0.8	0.1	-
U2 - dose-specific alpha	2.3	5.7	1.4	0.8	0.1	-
U3 - shared alpha	2.4	5.7	1.3	0.7	0.1	-
U3 - dose-specific alpha	2.4	5.7	1.3	0.7	0.1	-
Mean trial duration (days)						Days
U2 - shared alpha						940
U2 - dose-specific alpha						939
U3 - shared alpha						938.7
U3 - dose-specific alpha						937.9

Priors: efficacy beta-binomial Beta(0.5, 0.5); efficacy additive alpha Beta(0.15, 0.85); toxicity additive alpha Beta(0.15, 0.85).

Final column: early stopping % for MTD selection and no-OBD % for OBD selection; shaded rows are the dose-specific alpha efficacy model.